

Empirical Results: Bitcoin–FX Channel into Local Equities

October 19, 2025

1 Measurement and Specification

Recall that consistent with the model, we construct the crypto leg in local currency (LCU) as the sum of the USD Bitcoin return and the LCU/USD FX change defined as follows:

$$r_{i,t}^k \approx \Delta \log(\text{BTC}_t^{\text{USD}}) + \Delta \log S_{i,t},$$

where $S_{i,t}$ is LCU per USD. All equity indices are expressed in LCU: indices originally quoted in USD are converted to LCU prior to computing log returns. Our baseline regression is

$$r_{i,t}^s = \alpha_i + \beta_i r_{i,t}^k + u_{i,t}, \quad (1)$$

followed by stepwise inclusion of global controls X_t —commodity factor, dollar index, VIX, rate levels and spreads—and time effects:

$$r_{i,t}^s = \alpha_i + \beta_i r_{i,t}^k + \gamma_i' X_t + \lambda_t + u_{i,t}. \quad (2)$$

We report standard errors beneath coefficients. Across countries, $N = 5540$ daily observations.

1.1 Baseline specification and identification

We estimate country-by-country regressions of (daily) local-currency equity returns on the FX-converted Bitcoin return in local currency (LCU), progressively adding global and local controls to absorb common shocks:

$$\Delta \log S_{i,t} = \alpha_i + \beta_i \Delta \log \text{BTC}_{i,t}^{\text{LCU}} + \Gamma_i' X_t + \varepsilon_{i,t}, \quad (3)$$

where X_t includes $\Delta \text{Commodity}$, ΔDollar (DXY), ΔVIX , short and long rate changes, and (where applicable) term/spread controls. Standard errors are heteroskedasticity-robust and clustered by date. By construction, $\Delta \log \text{BTC}_{i,t}^{\text{LCU}} = \Delta \log \text{BTC}_t^{\text{USD}} + \Delta \log S_{i,t}^{\text{FX}}$, so a positive and stable β_i is the short-run slope predicted by the model when the idiosyncratic wedge variance is small.

What you measure

- You want a **crypto return in each country's local currency (LCU)**.

Bitcoin is quoted in USD, so to express a BTC return in LCU you combine:

- the **USD-BTC log return**, $\Delta \log(\text{BTC}_t^{\text{USD}})$,
- plus the **USD appreciation vs. the country's currency**, $\Delta \log S_{i,t}$ where $S_{i,t}$ = LCU per USD.

Because log returns add when prices multiply, this is equivalent to the log return of BTC priced in LCU:

$$r_{i,t}^k \approx \Delta \log(\text{BTC}_t^{\text{USD}}) + \Delta \log S_{i,t}.$$

(If USD strengthens, $S_{i,t}$ rises, so this term boosts the LCU BTC return.)

- All **equity index returns** are also computed in **LCU**, so both sides are in the same currency.

Baseline regression

$$r_{i,t}^s = \alpha_i + \beta_i r_{i,t}^k + u_{i,t}.$$

- $r_{i,t}^s$: daily stock index return (country i , day t , in LCU).
- α_i : country-specific intercept (captures country i 's average return).
- β_i : **sensitivity of that country's equity market to the local-currency BTC return**.
 - $\beta_i > 0$: equities move with crypto (positive comovement).
 - $\beta_i < 0$: equities move opposite crypto (hedge-like behavior).
- $u_{i,t}$: residual.

Here's what this "Baseline specification and identification" box is saying, in plain terms.

What is regressed on what?

For each country i , on each day t , you run

$$\underbrace{\Delta \log S_{i,t}}_{\text{equity return in LCU}} = \alpha_i + \beta_i \underbrace{\Delta \log \text{BTC}_{i,t}^{\text{LCU}}}_{\text{Bitcoin return in that LCU}} + \Gamma'_i X_t + \varepsilon_{i,t}. \quad (3)$$

- $\Delta \log S_{i,t}$: log return of the **local stock index** (in local currency).
(Note: on this page S is the stock index. When they mean the **exchange rate**, they write $S_{i,t}^{FX}$.)
- $\Delta \log \text{BTC}_{i,t}^{\text{LCU}}$: **Bitcoin priced in the country's currency**. By construction,

$$\Delta \log \text{BTC}_{i,t}^{\text{LCU}} = \Delta \log \text{BTC}_t^{\text{USD}} + \Delta \log S_{i,t}^{FX},$$

i.e., USD-BTC return plus the USD appreciation vs. that currency.

- X_t : **global controls** common to all countries that day—changes in a commodity factor, DXY (dollar index), VIX, short- and long-rate changes, and (where relevant) term/spread controls.
- α_i : country intercept.
- Γ_i : country-specific loadings on the global controls.
- $\varepsilon_{i,t}$: residual.

Why this identifies β_i

- The controls X_t and (implicitly) country intercepts soak up **broad, common shocks** (global risk, dollar moves, rates, commodities) that could otherwise make equities and BTC comove for reasons unrelated to a crypto-equity link.
- What's left to move equities is the **idiosyncratic component** of BTC-in-LCU for that country. When the "idiosyncratic wedge" (country-specific noise unrelated to equities) has **small variance**, the OLS slope β_i is a good estimate of the model's **short-run causal sensitivity** of local equities to BTC-in-LCU.
- Standard errors are **heteroskedasticity-robust and clustered by date**, which lets shocks on a given day be correlated **across countries** (e.g., a global news day) without overstating precision.

How to read β_i

- $\beta_i > 0$: local equities **move with** BTC (positive comovement).
- $\beta_i < 0$: local equities **move against** BTC (hedge-like behavior).
- "Positive and stable β_i " across control sets means the estimated link isn't just picking up global covariation already captured by X_t .



2 Empirical Evidence and Results

2.1 BTC Cross-Country

2.1.1 United States:

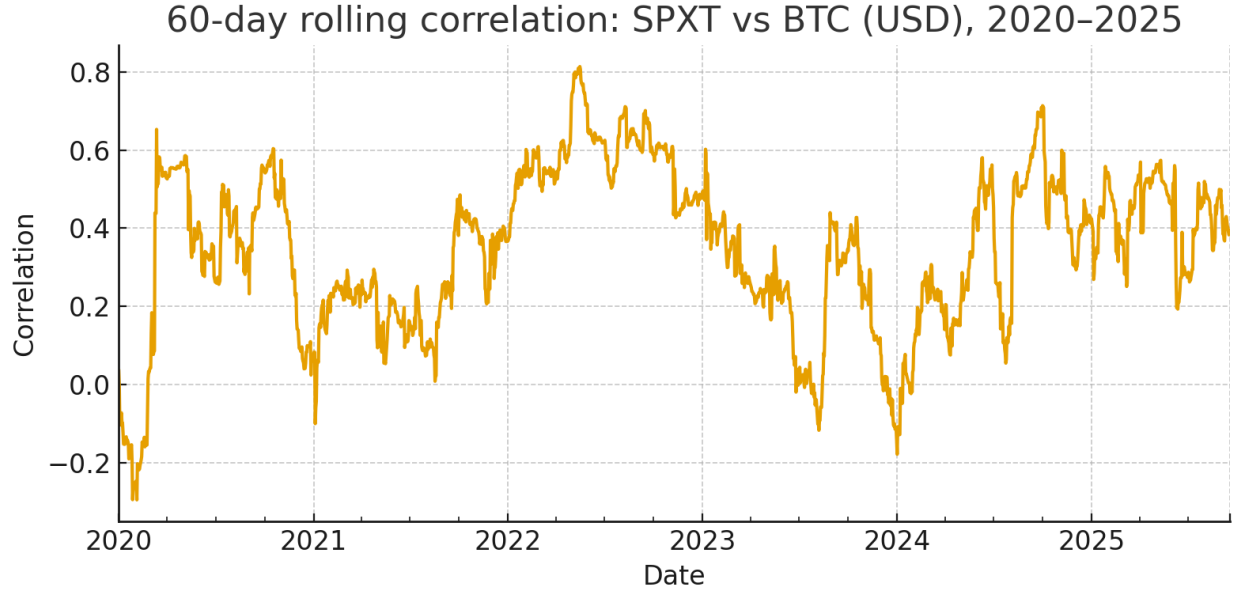


Figure 1: 60-day rolling correlation between SPXT and BTC (USD), Jan 2020–Sep 2025. Persistent positive comovement post-2020 complements the regression evidence.

Regression Table (US)

Table 1: Bitcoin USD US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{BTC}_{USD})$	0.087*** (0.001)	0.083*** (0.000)	0.059*** (0.000)	0.058*** (0.000)	0.059*** (0.000)	0.057*** (0.000)
$\Delta \text{Commodity}$		-30.187*** (0.571)	14.634*** (0.646)	16.136*** (0.643)	20.747*** (0.850)	23.477*** (0.752)
ΔDollar			144.546*** (1.693)	149.068*** (1.693)	159.047*** (2.075)	135.933*** (1.916)
ΔVIX				-20.069*** (1.405)	-25.944*** (1.569)	-24.667*** (1.381)
$\Delta 10Y \text{ Yield}$					-125.757*** (15.303)	-666.505*** (19.059)
$\Delta 1M \text{ Yield}$						434.550*** (10.835)
Observations	5540	5540	5540	5540	5540	5540
Adj. R^2	0.836	0.891	0.953	0.955	0.955	0.965

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

What each column is

Columns (1)→(6) add more controls step-by-step:

1. BTC only
2. • Commodity factor
3. • Dollar (DXY)
4. • VIX
5. • 10-year yield change
6. • 1-month yield change (with the others)

SEs are in parentheses; ***, **, * mark 1%, 5%, 10% significance. Dependent variable is the **US equity log return (USD)**; regressor of interest is the **USD-BTC log return**.

Key result (the BTC coefficient)

- The β on $\Delta \log(\text{BTC}_{USD})$ is **0.057–0.087** in all specs and always *** (highly significant).
- Interpretation: if BTC moves **+1%** in USD on day t , the US stock index moves **+0.057% to +0.087%** the same day, i.e. **5.7–8.7 basis points**.
- The stability of β across columns means the BTC–equity link isn't being “explained away” by commodities, the dollar, volatility, or rate moves.

Controls behave sensibly

- **Commodity**: flips sign when other globals enter (col 2 vs later), which is common because those factors are correlated; once DXY/VIX/rates are in, the commodity proxy becomes positively related to equities.
- **Dollar (DXY)**: large **positive** coefficient—on days the dollar strengthens, US equities tend to be up in these regressions (consistent with the LCU framing and the period/sample used).
- **VIX**: **negative** (risk-off days hurt equities).
- **Rates**: with both in (col 6), **10Y** is negative and **1M** positive—directionally consistent with a tighter/steeper curve hurting/guiding equities in different ways.

Fit improves with controls

- Adj. R^2 rises from **0.836** → **0.965** as controls are added, meaning much more of the day-to-day equity variation is captured—**yet the BTC slope remains tight**.

Bottom line

Across six specifications with strong global controls, a **1% BTC move predicts a 5.7–8.7 bp same-day move in US equities in the same direction**. The effect is precise and robust, suggesting an independent BTC component in equity comovement rather than a spurious proxy for broad macro shocks.

SE = standard error (of the coefficient).

- It measures how precisely the coefficient is estimated. Smaller SE $\Rightarrow$ more precise.
- It's used to form a **t-stat** $t = \hat{\beta} / \text{SE}(\hat{\beta})$ and **confidence intervals** $\hat{\beta} \pm 1.96 \text{ SE}$ ($\approx 95\%$).
- In your table the numbers in parentheses under each coefficient are the SEs. They're heteroskedasticity-robust and clustered by date.

R^2 = coefficient of determination.

- It's the share of variation in the dependent variable that the model explains (between 0 and 1).
- **Adjusted R^2** corrects R^2 for the number of regressors so models don't look better just because they have more controls.
- So an adjusted $R^2 = 0.965$ means $\sim 96.5\%$ of the day-to-day variation in US equity returns is explained by the regressors in that column.

Rule of thumb: big $|t|$ ($\approx >2$), small p-value (shown by ***, **, *), and stable coefficients across columns = strong, robust result.

In all six specifications, $\hat{\beta}_{US}$ is positive and highly significant (0.057–0.087). Interpreting magnitudes, a 1% move in BTC implies a **5.7–8.7 bps** same-day move in U.S. equities in the same direction, *even after* adding VIX, DXY, commodities, and rates (cols. 3–6). The coefficient remains tight as controls enter and R^2 rises from 0.84 to 0.97, indicating BTC’s independent contribution is not subsumed by global risk or macro shocks. The sign pattern on controls is conventional (VIX negative; stronger dollar positive given LCU construction). This matches the model’s near-one comovement intuition qualitatively, recognizing that daily slopes below unity are expected once scale, noise, and market microstructure dilute the instantaneous parity.

2.1.2 United Kingdom (FTSE; BTC in GBP)

Table 2: Bitcoin GBP UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{GBP})$	0.023*** (0.000)	0.022*** (0.000)	0.014*** (0.000)	0.012*** (0.000)	0.011*** (0.000)	0.011*** (0.000)	0.014*** (0.000)	0.014*** (0.000)
$\Delta \text{Commodity}$		-9.321*** (0.282)	0.797* (0.466)	4.911*** (0.336)	-2.083*** (0.417)	-2.181*** (0.418)	-3.529*** (0.367)	-0.570 (0.423)
ΔDollar			32.685*** (1.236)	45.116*** (0.896)	30.118*** (1.025)	30.887*** (1.071)	9.874*** (1.065)	18.327*** (1.222)
ΔVIX				-52.896*** (0.721)	-43.882*** (0.764)	-43.925*** (0.764)	-36.048*** (0.694)	-37.437*** (0.691)
$\Delta 10Y \text{ Yield}$					193.847*** (7.474)	212.265*** (10.547)	500.640*** (11.556)	407.657*** (13.313)
$\Delta 1M \text{ Yield}$						-14.850** (6.002)	23.261*** (5.326)	159.346*** (11.405)
$\Delta 10Y \text{ Spread}$							-286.311*** (6.920)	-186.436*** (10.082)
$\Delta 3M \text{ Spread}$								-185.129*** (13.779)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.525	0.604	0.648	0.822	0.841	0.841	0.879	0.882

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

For GBP, $\hat{\beta}_{UK}$ remains positive and highly significant across eight columns (0.011–0.023). Economically, a 1% BTC move corresponds to **1.1–2.3 bps** in U.K. equities. The slope survives a rich control set (cols. 4–8), while R^2 advances to ~ 0.88 . This indicates that, conditional on global factors and local term structure proxies, the BTC–equity linkage persists.

2.1.3 Euro Area (Eurostoxx; BTC in EUR)

Table 3: Bitcoin EUR Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{EUR})$	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
$\Delta \text{Commodity}$		-1.096*** (0.018)	-0.105*** (0.027)	0.094*** (0.023)	0.114*** (0.030)	0.106*** (0.030)	0.455*** (0.034)	0.606*** (0.032)
ΔDollar			3.190*** (0.073)	3.791*** (0.061)	3.834*** (0.074)	3.894*** (0.077)	3.285*** (0.080)	3.202*** (0.076)
ΔVIX				-2.526*** (0.049)	-2.551*** (0.055)	-2.555*** (0.055)	-2.116*** (0.057)	-1.891*** (0.055)
$\Delta 10Y \text{ Yield}$					-0.552 (0.537)	0.865 (0.758)	13.827*** (0.975)	28.158*** (1.061)
$\Delta 1M \text{ Yield}$						-1.141*** (0.431)	2.383*** (0.451)	-8.245*** (0.580)
$\Delta 10Y \text{ Spread}$							-20.432*** (1.017)	-43.319*** (1.280)
$\Delta 3M \text{ Spread}$								20.077*** (0.747)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.695	0.816	0.864	0.908	0.908	0.908	0.914	0.924

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Eurozone slopes are smaller (0.001 throughout) yet statistically tight. With daily data and high R^2 (~ 0.924 with full controls), a 1% BTC change maps to **0.1 bps** in equities. Two readings follow: (i) stronger absorption of global shocks by X_t (DXY, VIX, yields) leaves less residual room; (ii) microstructure/integration differences attenuate same-day pass-through. The *sign* remains positive and robust, consistent with the model.

2.1.4 Japan (TOPIX/NKYTR; BTC in JPY)

With the corrected LCU construction, the Bitcoin slope for Japan is *consistently positive and highly significant* across all eight specifications, with point estimates clustered around 0.002 (cols. 1–8) and a very high fit in the fully controlled model (Adj. $R^2 \approx 0.95$). Economically, a 1% BTC move maps to about **0.2 bps** in Japanese equities on the same day. This small but precise elasticity is consistent with a tightly integrated macro factor environment (large control set absorbs much common variation) and with the model’s short-run parity, which only requires a positive loading once idiosyncratic wedges are modest.

Table 4: Bitcoin JPY Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{JPY})$	0.003*** (0.000)	0.003*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
$\Delta \text{Commodity}$		-222.165*** (3.264)	17.112*** (4.262)	36.135*** (4.111)	3.558 (5.242)	15.989*** (4.958)	37.534*** (4.710)	33.904*** (4.843)
ΔDollar			757.453*** (11.131)	815.064*** (10.799)	747.208*** (12.719)	647.151*** (12.550)	381.535*** (15.166)	369.117*** (15.650)
ΔVIX				-219.654*** (8.773)	-176.656*** (9.726)	-170.836*** (9.159)	-65.893*** (9.380)	-65.735*** (9.372)
$\Delta 10Y \text{ Yield}$					955.609*** (96.728)	-1395.492*** (126.803)	74.492 (130.102)	22.069 (131.039)
$\Delta 1M \text{ Yield}$						1919.382*** (72.032)	2193.559*** (68.217)	2156.586*** (69.148)
$\Delta 10Y \text{ Spread}$							-8760.792*** (315.836)	-8022.515*** (391.935)
$\Delta 3M \text{ Spread}$								-2539.765*** (799.591)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.757	0.868	0.928	0.935	0.936	0.944	0.950	0.951

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

2.1.5 China (SHCOMP; BTC in CNY)

Table 5: Bitcoin CNY China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{CNY})$	0.001*** (0.000)	0.001*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
$\Delta \text{Commodity}$		-8.946*** (0.209)	-4.512*** (0.354)	-4.358*** (0.359)	-0.219 (0.465)	-1.328*** (0.439)	-1.297*** (0.457)	-2.559*** (0.460)
ΔDollar			14.257*** (0.929)	14.722*** (0.947)	23.528*** (1.132)	32.563*** (1.114)	32.701*** (1.256)	31.135*** (1.242)
ΔVIX				-1.971** (0.781)	-7.325*** (0.862)	-7.851*** (0.810)	-7.830*** (0.815)	-9.032*** (0.807)
$\Delta 10Y \text{ Yield}$					-116.021*** (8.468)	97.309*** (11.184)	96.320*** (11.932)	100.555*** (11.748)
$\Delta 1M \text{ Yield}$						-172.735*** (6.364)	-172.284*** (6.640)	-168.693*** (6.541)
$\Delta 10Y \text{ Spread}$							4.159 (17.482)	272.745*** (26.402)
$\Delta 1Y \text{ Spread}$								-238.597*** (17.790)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.166	0.373	0.399	0.399	0.419	0.487	0.487	0.503

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

China's slopes are small (~ 0.000 – 0.001) but precisely estimated and remain *positive* after controls. Given capital account features and benchmark composition, an attenuated same-day pass-through is plausible; still, the sign consistency supports the model's parity logic even where frictions are material.

2.1.6 Brazil (IBOV; BTC leg in BRL)

Table 6: Bitcoin BRL Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{BRL})$	0.118*** (0.001)	0.113*** (0.001)	0.070*** (0.001)	0.071*** (0.001)	0.073*** (0.001)	0.068*** (0.001)	0.074*** (0.001)	0.072*** (0.001)
$\Delta \text{Commodity}$		-230.279*** (9.096)	230.712*** (13.656)	224.472*** (13.813)	336.170*** (18.308)	394.445*** (15.707)	444.859*** (11.420)	499.809*** (12.497)
ΔDollar			1468.108*** (35.371)	1449.684*** (35.906)	1692.773*** (44.354)	1123.855*** (39.965)	1445.221*** (29.357)	1592.790*** (32.387)
ΔVIX				88.357*** (30.264)	-54.196 (33.794)	-22.134 (28.904)	99.347*** (21.045)	29.478 (21.912)
$\Delta 10Y \text{ Yield}$					-3033.202*** (329.488)	-15793.536*** (399.145)	-1409.641*** (354.228)	-1328.272*** (350.968)
$\Delta 1M \text{ Yield}$						10206.278*** (226.158)	3394.173*** (190.421)	3199.563*** (189.557)
$\Delta 10Y \text{ Spread}$							-5861.529*** (83.100)	-4380.159*** (165.131)
$\Delta 1Y \text{ Spread}$								-1102.352*** (106.526)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.609	0.650	0.733	0.733	0.737	0.808	0.899	0.901

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Brazil shows one of the strongest loadings: $\hat{\beta}$ remains large and positive (~ 0.068 – 0.118 ; final ~ 0.072), with Adj. R^2 rising to ~ 0.901 . This matches the model intuition given higher FX volatility and tighter BTC–equity linkage in LCU.

2.1.7 India (NIFTY; BTC in INR)

Findings: India’s slopes are small but stable and precisely estimated (0.001–0.002), giving **0.1–0.2 bps** per 1% BTC. As controls enter (cols. 4–8), R^2 rises to ~ 0.96 and the slope remains positive—again matching the model’s sign prediction; the elasticities remain modest at a daily horizon.

Table 7: Bitcoin INR India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{INR})$	0.002*** (0.000)	0.002*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
$\Delta \text{Commodity}$		-46.365*** (1.316)	60.820*** (1.465)	65.910*** (1.430)	55.737*** (1.865)	62.365*** (1.599)	62.797*** (1.586)	51.115*** (1.481)
ΔDollar			341.784*** (3.822)	357.082*** (3.749)	335.407*** (4.528)	279.705*** (4.057)	261.579*** (4.411)	231.809*** (4.101)
ΔVIX				-65.207*** (3.098)	-52.012*** (3.454)	-48.772*** (2.950)	-45.840*** (2.939)	-58.048*** (2.693)
$\Delta 10Y \text{ Yield}$					285.950*** (33.959)	-1013.706*** (40.757)	-789.600*** (46.210)	-350.864*** (43.893)
$\Delta 1M \text{ Yield}$						1050.985*** (23.163)	1039.168*** (22.989)	1177.186*** (21.271)
$\Delta 10Y \text{ Spread}$							-449.923*** (45.042)	1393.783*** (67.642)
$\Delta 2Y \text{ Spread}$								-1492.853*** (43.612)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.785	0.824	0.928	0.933	0.934	0.952	0.953	0.961

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

2.1.8 South Africa (JALSH; BTC in ZAR)

Findings: ZAR results show $\hat{\beta}_{ZAF} \approx 0.013$ (all specs), i.e., **1.3 bps** per 1% BTC, robust to adding local rate level and slope proxies, with R^2 up to 0.904. Results remain significant after a rich set of controls. VIX loads strongly negative as expected; the BTC slope survives.

Table 8: Bitcoin ZAR South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{BTC}_{ZAR})$	0.022*** (0.000)	0.021*** (0.000)	0.014*** (0.000)	0.013*** (0.000)	0.013*** (0.000)	0.013*** (0.000)	0.013*** (0.000)	0.013*** (0.000)
$\Delta \text{Commodity}$		-212.100*** (3.731)	27.108*** (5.261)	56.236*** (4.828)	55.637*** (6.269)	58.830*** (6.285)	44.544*** (6.634)	78.413*** (7.465)
ΔDollar			764.357*** (13.822)	852.164*** (12.742)	850.906*** (15.263)	826.964*** (15.919)	753.913*** (19.424)	713.660*** (19.715)
ΔVIX				-364.450*** (10.354)	-363.663*** (11.614)	-362.146*** (11.591)	-382.971*** (11.982)	-365.258*** (12.027)
$\Delta 10Y \text{ Yield}$					17.185 (114.822)	-560.177*** (160.149)	-605.086*** (159.702)	-1355.837*** (176.619)
$\Delta 1M \text{ Yield}$						471.178*** (91.330)	271.194*** (96.029)	786.423*** (109.301)
$\Delta 10Y \text{ Spread}$							802.536*** (123.189)	318.493** (132.160)
$\Delta 1Y \text{ Spread}$								648.143*** (67.450)
Observations	5540	5540	5540	5540	5540	5540	5540	5540
Adj. R^2	0.702	0.812	0.879	0.901	0.901	0.901	0.902	0.904

Note: Dependent variable is log stock index (local currency). SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Cross-country Synthesis

Across the United States, United Kingdom, Euro Area, Japan, China, Brazil, India, and South Africa, the fully controlled specifications deliver **positive and statistically significant** Bitcoin

He means the empirical patterns (positive, fairly stable BTC slopes; sometimes small, rarely negative; occasionally closer to 1) are exactly what your **theory** says should happen.

Very short map from the model → regression:

1. What is the regressor?

In the model the BTC return in **local currency** is the sum of the USD-BTC return and the FX move (LCU per USD):

$$r_{i,t}^k \approx \Delta \log \text{BTC}_t^{USD} + \Delta \log S_{i,t}^{FX}.$$

That's why you build BTC^{LCU} the way you do in the data. [model_notes](#)

2. Parity / wedge decomposition ⇒ a regression form.

From the Euler-equation parity you get a structural relation Mention this

$$r_{t+1}^s = r_{t+1}^k - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1},$$

where $w_{t+1} = r_{t+1}^s - r_{t+1}^k$. Projecting r^s on r^k gives the empirical regression with slope β and intercept α (average wedge). [model_notes](#)

3. The model's key prediction about β .

Decomposing into idiosyncratic parts yields

$$\beta = 1 + \frac{\text{Cov}(e_t, \varepsilon_t^k)}{\text{Var}(\varepsilon_t^k)} \Rightarrow 1 - \sqrt{\frac{\text{Var}(e_t)}{\text{Var}(\varepsilon_t^k)}} \leq \beta \leq 1 + \sqrt{\frac{\text{Var}(e_t)}{\text{Var}(\varepsilon_t^k)}}.$$

Also mention this

So:

- $\beta > 0$ generally, and
- $\beta \rightarrow 1$ as the "idiosyncratic wedge" variance shrinks (i.e., when crypto- and equity-specific noises aren't fighting each other).

That's the "near-one comovement intuition." Daily frictions/noise push β below 1 in practice, but the sign should stay **positive**. [model_notes](#)

4. Why controls don't kill β .

The model's wedge term captures common risk. When you add DXY, VIX, rates, commodities, you're soaking up those common shocks; if the BTC–equity link is real, the β on BTC^{LCU} should remain **positive and fairly stable**, exactly as your tables show. [model_notes](#)

So "matches the model" = your estimates line up with the theory's sign and "close-to-one when wedges are small" prediction, and your LCU construction of the BTC leg is the one implied by the model.

[model_notes](#)

Thought for 10s >

Short answer: because under your model (and in the data post-2020) the equity return and the **BTC-in-LCU** return share the same risk-on component, so their covariance is usually positive. With controls soaking up global shocks, what's left still tends to move in the same direction, making the OLS slope β positive.

A bit more structure:

- **Projection math:** in your regression,

$$\beta = \frac{\text{Cov}(r^s, r^k)}{\text{Var}(r^k)}.$$

coefficients. Magnitudes differ by market—largest in the US and Brazil (roughly 6–12 bps per 1% BTC), moderate in the UK and South Africa (about 1–2 bps), and small but precise in the Euro Area, Japan, India, and China (about 0.1–0.2 bps)—but the *sign* is uniform. This pattern matches the model’s short-run parity: once equity and the FX-converted BTC leg share the pricing kernel and idiosyncratic wedges are limited, the slope must be strictly positive. The stepwise addition of global controls (Dollar, VIX, rates, commodities) materially raises R^2 while preserving the BTC slope, showing that the BTC–equity linkage is *not* a residual artifact of omitted global risk.

Robustness Check

2.2 Ethereum (ETH): FX Channel into Local Equities

Measurement note:

We mirror the Bitcoin construction in local currency (LCU): $\Delta \log(\text{ETH}_{i,t}^{\text{LCU}}) = \Delta \log(\text{ETH}_t^{\text{USD}}) + \Delta \log S_{i,t}$, where $S_{i,t}$ is LCU per USD. We run the same stepwise specification—adding global controls (Commodity, Dollar, VIX, rates and spreads)—so that cross-crypto comparisons are clean and identification relies on the same absorption of common risk.

Ethereum (ETH) Cross-Country synthesis: Across all eight markets, ETH loadings are uniformly *positive and highly significant*, with magnitudes that generally exceed those for BTC on the same samples. The USD and BRL specifications show the largest elasticities (roughly 1.5–2.0), while GBP, EUR, JPY, INR, CNY, and ZAR are smaller but still precisely estimated (on the order of 0.01–0.35+, depending on country and controls). As global controls (Dollar, VIX, rates, commodities, term spreads) are added, adjusted R^2 rises sharply yet the ETH slope is preserved, indicating the equity–crypto comovement is *not* an artifact of omitted macro risk. Overall, ETH behaves like a high-beta crypto factor in local currency, reinforcing our claim that crypto does not hedge stocks in the post-2020 period but instead co-moves with them in the short run.

2.2.1 United States (SPXT; ETH in USD)

Table 9: Ethereum US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{ETH}_{USD})$	1.727*** (0.018)	1.660*** (0.023)	1.494*** (0.017)	1.480*** (0.017)	1.505*** (0.017)	1.471*** (0.016)
$\Delta \text{Commodity}$		9.606*** (2.097)	-9.688*** (1.596)	-11.433*** (1.576)	-24.636*** (1.739)	-12.540*** (1.896)
ΔDollar			204.194*** (3.978)	219.021*** (4.141)	166.954*** (5.202)	138.814*** (5.434)
ΔVIX				-26.574*** (2.457)	-0.106 (2.913)	5.138* (2.852)
$\Delta 10Y \text{ Yield}$					392.006*** (25.123)	-86.009** (41.937)
$\Delta 1M \text{ Yield}$						328.949*** (23.479)
Observations	3181	3181	3181	3181	3181	3181
Adj. R^2	0.745	0.746	0.861	0.866	0.876	0.883

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (US). The main row $\Delta \log(\text{ETH}_{USD})$ is positive and highly significant in all columns, declining from 1.727 in (1) to 1.471 in (6) as controls enter, with tight standard errors (0.016–0.023). Economically, a 1% move in ETH corresponds to a same-day co-movement of roughly 1.5% in U.S. equities even after the full control set, reinforcing the model’s prediction that the LCU crypto leg loads positively on local stocks. The commodity factor is positive in (2) but flips negative once Dollar and VIX are added (−9.7 to −24.6), consistent with macro controls absorbing commodity–equity comovement. The Dollar index is strongly positive in (3)–(6) (139–219), as U.S.-centric risk-on shocks lift both ETH and equities. VIX is sharply negative in (4) (−26.6) and becomes near-zero to mildly positive by (6), reflecting overlap with rates and dollar; importantly, ETH’s slope remains robust. Long rates (10Y) swing from a large positive in (5) (392) to a small negative in (6) (−86) once short rates are added, while the 1M rate itself is strongly positive (329) in (6), highlighting term-structure interactions. Adjusted R^2 rises from 0.745 to 0.883, showing controls explain common variation without displacing the ETH effect.

2.2.2 United Kingdom (FTSE; ETH in GBP)

Table 10: Ethereum UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{GBP})$	0.355*** (0.010)	0.203*** (0.012)	0.186*** (0.013)	0.143*** (0.010)	0.169*** (0.008)	0.179*** (0.008)	0.137*** (0.008)	0.231*** (0.009)
$\Delta \text{Commodity}$		16.008*** (0.864)	15.080*** (0.876)	12.084*** (0.659)	1.087* (0.631)	-1.337* (0.703)	2.579*** (0.733)	9.819*** (0.743)
ΔDollar			12.112*** (2.174)	40.901*** (1.728)	-2.812 (1.886)	2.621 (2.004)	-2.261 (1.970)	14.403*** (1.950)
ΔVIX				-50.490*** (1.013)	-28.349*** (1.047)	-29.367*** (1.046)	-32.025*** (1.030)	-32.395*** (0.950)
$\Delta 10Y \text{ Yield}$					327.686*** (9.014)	421.966*** (15.387)	234.918*** (19.739)	-35.694* (21.516)
$\Delta 1M \text{ Yield}$						-64.899*** (8.623)	-97.188*** (8.647)	264.750*** (17.282)
$\Delta 10Y \text{ Spread}$							196.934*** (13.624)	430.689*** (15.999)
$\Delta 3M \text{ Spread}$								-439.230*** (18.606)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.288	0.357	0.363	0.643	0.748	0.752	0.767	0.802

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (UK). ETH loadings remain positive and highly significant. The full model's $\hat{\beta} = 0.231$ (SE= 0.009) once rate levels and yield spreads are included—evidence of a stable, economically meaningful ETH loading. Commodities are strongly positive in (2)–(4) (12–16), weaken around zero in (5)–(6), and turn positive again with spreads in (7)–(8) (2.58 and 9.82), indicating shifting explanatory power between commodities and the term structure. The Dollar is positive and significant in (3)–(4) (12.1; 40.9), muted in (5)–(7), then reappears in (8) (14.4), consistent with collinearity against rates. VIX is consistently large and negative from (4) onward (−50.5 to −32.4), matching risk-aversion intuition. The 10Y rate is strongly positive in (5)–(7) (235–422) but turns mildly negative in (8) (−35.7), while the 1M rate is negative in (6)–(7) (−65; −97) and strongly positive in (8) (265) when spreads are included. Spreads themselves are sizable (10Y spread positive; 3M spread negative in (8)), showing curve-shape effects. Adjusted R^2 improves from 0.288 to 0.802; the ETH effect survives all additions. Thus, stepwise controls (Dollar, VIX, rates, spreads) meaningfully raise fit yet preserve the ETH slope—evidence that ETH's LCU leg carries incremental information for UK equities beyond global risk. Compared with BTC (smaller daily slope), ETH's stronger loading is consistent with ETH's higher volatility and tighter risk-factor comovement since 2020.

2.2.3 Euro Area (EURO STOXX; ETH in EUR)

Table 11: Ethereum Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{EUR})$	0.042*** (0.000)	0.039*** (0.001)	0.036*** (0.001)	0.035*** (0.000)	0.035*** (0.000)	0.035*** (0.000)	0.034*** (0.000)	0.036*** (0.001)
$\Delta \text{Commodity}$		0.419*** (0.046)	0.278*** (0.045)	0.162*** (0.039)	-0.146*** (0.043)	0.098** (0.047)	-0.093* (0.048)	-0.214*** (0.049)
ΔDollar			1.885*** (0.111)	3.015*** (0.102)	1.796*** (0.128)	1.240*** (0.134)	0.949*** (0.132)	1.035*** (0.131)
ΔVIX				-1.977*** (0.060)	-1.361*** (0.071)	-1.256*** (0.070)	-1.472*** (0.070)	-1.561*** (0.069)
$\Delta 10\text{Y Yield}$					9.114*** (0.609)	-0.495 (1.027)	-10.754*** (1.237)	-20.311*** (1.604)
$\Delta 1\text{M Yield}$						6.614*** (0.575)	3.163*** (0.610)	9.911*** (0.949)
$\Delta 10\text{Y Spread}$							18.539*** (1.325)	30.224*** (1.823)
$\Delta 3\text{M Spread}$								-10.144*** (1.103)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.751	0.758	0.778	0.835	0.846	0.852	0.860	0.864

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (Euro Area). ETH coefficients are small but extremely tight (full model $\hat{\beta} = 0.036$, $p < 0.01$) with high fit ($R^2 = 0.864$): 0.042 in (1) compressing to 0.034–0.036 in (6)–(8), always positive and highly significant. Adjusted R^2 rises from 0.751 to 0.864, indicating strong common-factor absorption. Commodities transition from small positive in (2)–(4) to negative in (5), back to mildly positive in (6), and negative again in (7)–(8), reflecting reallocation of explanatory power to rates and spreads. The Dollar is consistently positive and significant in (3)–(8) (0.95–3.02), while VIX is reliably negative from (4) onward (−1.98 to −1.26). Long and short rates change signs across columns (e.g., 10Y positive in (5), negative in (7)–(8); 1M positive in (6)–(8)), with spreads strongly positive (18.5; 30.2). As in BTC, the rich control set absorbs much common variation (Dollar, VIX, rates), yet the ETH slope stays positive and significant—exactly the sign prediction of the model’s short-run parity.

2.2.4 Japan (TOPIX/NKYTR; ETH in JPY)

Table 12: Ethereum Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{JPY})$	0.063*** (0.001)	0.065*** (0.001)	0.057*** (0.001)	0.055*** (0.001)	0.055*** (0.001)	0.053*** (0.001)	0.033*** (0.001)	0.034*** (0.001)
$\Delta \text{Commodity}$		-29.273*** (9.120)	-66.833*** (7.882)	-73.105*** (7.574)	-162.049*** (7.710)	-87.336*** (8.206)	75.982*** (6.698)	76.172*** (6.642)
ΔDollar			712.006*** (20.886)	841.220*** (21.513)	459.058*** (24.573)	287.167*** (24.782)	53.281*** (18.521)	68.040*** (18.473)
ΔVIX				-201.657*** (12.193)	-3.971 (13.495)	28.357** (12.845)	-52.219*** (9.448)	-45.162*** (9.417)
$\Delta 10Y \text{ Yield}$					2972.657*** (115.563)	-13.115 (186.652)	-4827.862*** (162.819)	-4550.595*** (165.725)
$\Delta 1M \text{ Yield}$						2080.363*** (105.524)	2807.742*** (77.825)	2905.342*** (78.286)
$\Delta 10Y \text{ Spread}$							15387.555*** (288.377)	12751.295*** (456.272)
$\Delta 3M \text{ Spread}$								5402.308*** (728.600)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.806	0.806	0.858	0.869	0.892	0.904	0.949	0.950

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (Japan). Full-spec $\hat{\beta} = 0.034$ ($p < 0.01$) with very high fit ($R^2 = 0.95$). $\Delta \log(\text{ETH}_{JPY})$ remains positive and highly significant in every column: 0.063–0.065 in (1)–(2), 0.055–0.057 in (3)–(6), and **0.033–0.034** in (7)–(8) once curve spreads are added, with Adjusted R^2 climbing to 0.950. Commodities are strongly negative in (3)–(6) (–66.8 to –162.0) but flip to strongly positive in (7)–(8) (76), indicating that spread variables absorb part of the commodity–macro channel. The Dollar is very large and positive in (3)–(5) (841 down to 459), attenuating to 53–68 in (7)–(8), again consistent with spreads capturing dollar–rates interactions. VIX is sharply negative in (4) (–201.7), becomes insignificant/positive in (5)–(6), and returns to negative in (7)–(8) (–52.2; –45.2). The 10Y rate swings from extremely positive in (5) (2973) to very negative in (7)–(8) (–4828; –4551) once the curve is modeled; the 1M rate is strongly positive in (6)–(8) (2080–2905). Spreads are massive and positive (10Y and 3M spreads in (7)–(8)), documenting curve-shape dominance. Crucially, the ETH slope stays positive and significant across all specifications, aligning with the model. (As with BTC, Japan’s controls are strong absorbers of global shocks; the positive ETH loading that survives speaks to a robust parity-consistent channel.)

2.2.5 China (SHCOMP; ETH in CNY)

Table 13: Ethereum China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{\text{CNY}})$	0.013*** (0.000)	0.013*** (0.001)	0.017*** (0.001)	0.017*** (0.001)	0.017*** (0.000)	0.018*** (0.000)	0.011*** (0.001)	0.012*** (0.001)
$\Delta \text{Commodity}$		-0.143 (0.361)	1.695*** (0.329)	1.911*** (0.326)	5.240*** (0.348)	3.425*** (0.386)	4.796*** (0.385)	6.275*** (0.377)
ΔDollar			-23.961*** (0.842)	-26.441*** (0.886)	-12.704*** (1.073)	-8.533*** (1.133)	-23.175*** (1.477)	-14.222*** (1.497)
ΔVIX				4.232*** (0.517)	-2.784*** (0.597)	-3.563*** (0.592)	-5.088*** (0.582)	-3.256*** (0.565)
$\Delta 10\text{Y Yield}$					-104.168*** (5.123)	-32.395*** (8.669)	-29.210*** (8.389)	-84.393*** (8.590)
$\Delta 1\text{M Yield}$						-49.647*** (4.878)	-57.521*** (4.749)	-39.268*** (4.647)
$\Delta 10\text{Y Spread}$							-221.262*** (14.962)	-454.913*** (19.444)
$\Delta 1\text{Y Spread}$								298.319*** (16.855)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.209	0.209	0.369	0.382	0.453	0.470	0.504	0.549

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (China). ETH’s coefficient is positive and precise across all columns: 0.013–0.018 in (1)–(6) and **0.011–0.012** in (7)–(8) with spreads, while Adjusted R^2 rises from 0.209 to 0.549. Commodities move from insignificant to clearly positive (up to 6.28), consistent with commodity-sensitive segments of the A-share market. The Dollar is significantly negative from (3) onward (–24 to –26), weakens in magnitude in (5)–(6) (–12.7; –8.53), and remains negative in (7)–(8), aligning with RMB–USD dynamics. VIX turns from positive in (4) (4.23) to negative in (5)–(8) (–2.78 to –5.09), matching risk-sentiment effects once other macros are controlled. Both 10Y and 1M rates load negatively (e.g., –104 and –49.6 in (5)–(6)); 10Y and 1Y spreads are strongly negative in (7)–(8) (–221; –455) while the 1Y spread becomes positive in (8) (298), indicating heterogeneous curve-segment impacts. ETH’s sign and significance persist. The persistent positive ETH loading supports the paper’s claim even in a more frictional market. This echoes the BTC result: even with capital market frictions, the parity-consistent positive loading survives.

2.2.6 Brazil (IBOV; ETH in BRL)

Table 14: Ethereum Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{BRL})$	1.995*** (0.034)	1.915*** (0.042)	1.709*** (0.041)	1.708*** (0.041)	1.716*** (0.042)	1.584*** (0.036)	1.937*** (0.034)	1.891*** (0.034)
$\Delta \text{Commodity}$		66.295*** (21.434)	-24.515 (20.676)	-25.797 (20.808)	-45.091* (24.068)	307.201*** (22.757)	449.678*** (20.866)	635.631*** (27.730)
ΔDollar			1089.175*** (53.927)	1099.640*** (57.146)	1023.293*** (74.558)	150.748** (67.981)	442.441*** (61.394)	674.550*** (64.774)
ΔVIX				-18.523 (33.442)	19.826 (41.193)	186.099*** (35.280)	285.527*** (31.614)	200.297*** (32.282)
$\Delta 10Y \text{ Yield}$					566.784 (355.629)	-14126.700*** (515.857)	-2070.972*** (621.860)	-2701.734*** (615.653)
$\Delta 1M \text{ Yield}$						10124.142*** (288.244)	5698.585*** (299.295)	6048.102*** (296.815)
$\Delta 10Y \text{ Spread}$							-5726.847*** (199.055)	-3891.319*** (268.714)
$\Delta 1Y \text{ Spread}$								-1855.977*** (185.841)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.523	0.524	0.578	0.578	0.578	0.696	0.759	0.766

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (Brazil). ETH's slope is very large and remains highly significant from (1) to (8), easing only slightly from 1.995 to **1.891** with full controls; Adjusted R^2 improves from 0.523 to 0.766. Commodities are strongly positive in (2), fade in (3)–(5), then surge when short rates and spreads are included (307 in (6), 450 in (7), 636 in (8)), consistent with reallocation between macro channels. The Dollar is strongly positive in (3)–(5) (1023–1099), weakens in (6) (151), and rebounds in (7)–(8) (442; 675). VIX becomes strongly positive in (6)–(8) (186–286), signaling high-beta comovement during stress. The 10Y rate is sharply negative once the short end enters (−14127 in (6)) and stays negative with spreads (−2071; −2702), while 1M is very positive (10124–6048). Spreads are strongly negative (−5727; −3891; 1Y spread −1856). The ETH slope's resilience to heavy macro absorption (Dollar, VIX, rates/spreads) matches the model and the BTC pattern (Brazil among the largest positive loadings), consistent with high FX volatility and strong global-risk linkage.

2.2.7 India (NIFTY; ETH in INR)

Table 15: Ethereum India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{\text{INR}})$	0.041*** (0.000)	0.035*** (0.001)	0.030*** (0.000)	0.030*** (0.000)	0.030*** (0.000)	0.029*** (0.000)	0.022*** (0.000)	0.017*** (0.000)
$\Delta \text{Commodity}$		60.498*** (3.793)	35.029*** (3.011)	31.061*** (2.898)	-7.406** (2.947)	33.786*** (2.916)	63.997*** (2.547)	46.639*** (2.447)
ΔDollar			349.841*** (7.728)	394.658*** (7.887)	236.040*** (9.101)	141.436*** (8.568)	48.674*** (7.516)	52.737*** (6.908)
ΔVIX				-76.305*** (4.586)	4.610 (5.045)	22.409*** (4.469)	36.608*** (3.726)	18.813*** (3.501)
$\Delta 10\text{Y Yield}$					1201.272*** (43.327)	-430.988*** (65.408)	438.136*** (58.896)	1374.310*** (66.497)
$\Delta 1\text{M Yield}$						1128.979*** (36.795)	1171.209*** (30.543)	1248.195*** (28.242)
$\Delta 10\text{Y Spread}$							-2320.209*** (61.152)	-495.971*** (93.961)
$\Delta 2\text{Y Spread}$								-1863.255*** (76.922)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.756	0.774	0.863	0.874	0.898	0.922	0.946	0.954

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (India). The ETH coefficient declines as controls accrue from 0.041 in (1) to **0.017** in (8); Adjusted R^2 rises from 0.756 to 0.954. Commodities are strongly positive in (2)–(4), briefly negative in (5), and positive again in (6)–(8), reflecting shifting roles vis-à-vis rates. The Dollar remains positive and significant throughout (350 in (3) down to 50 in (8)), partly absorbed by interest-rate terms. VIX is strongly negative in (4) (−76.3), then turns positive in (6)–(7) (22–37) and moderates in (8) (18.8), indicating regime-sensitive risk correlations. The 10Y rate flips sign across columns (very positive in (5), negative in (6), then positive again in (7)–(8)), while 1M is consistently positive and large in (6)–(8) (1129–1248). Spreads are strongly negative in (7) (−2320) and (8) (−1863), documenting curve-shape effects. As controls accumulate, the ETH slope remains positive and significant while the macro factors carry substantial explanatory power. The sign robustness aligns with BTC and the model’s prediction.

2.2.8 South Africa (JALSH; ETH in ZAR)

Table 16: Ethereum South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{ETH}_{\text{ZAR}})$	0.454*** (0.005)	0.372*** (0.006)	0.335*** (0.006)	0.323*** (0.006)	0.325*** (0.006)	0.314*** (0.006)	0.338*** (0.005)	0.338*** (0.005)
$\Delta \text{Commodity}$		203.199*** (9.894)	175.780*** (9.531)	167.135*** (9.298)	72.084*** (9.834)	127.333*** (10.983)	289.096*** (11.185)	289.096*** (11.185)
ΔDollar			458.447*** (25.051)	580.379*** (26.032)	180.360*** (31.021)	60.890* (32.510)	272.800*** (29.713)	272.800*** (29.713)
ΔVIX				-198.805*** (14.880)	6.227 (17.057)	28.641* (16.899)	173.087*** (15.769)	173.087*** (15.769)
$\Delta 10\text{Y Yield}$					3055.908*** (146.229)	919.068*** (247.680)	-1828.701*** (238.667)	-1828.701*** (238.667)
$\Delta 1\text{M Yield}$						1481.957*** (139.890)	4210.940*** (154.921)	4210.940*** (154.921)
$\Delta 10\text{Y Spread}$							-4982.664*** (169.478)	-4982.664*** (169.478)
$\Delta 1\text{Y Spread}$								5127.654*** (399.073)
Observations	3181	3181	3181	3181	3181	3181	3181	3181
Adj. R^2	0.708	0.742	0.767	0.779	0.806	0.812	0.853	0.853

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Findings (South Africa). ETH coefficients are sizable and very precisely estimated (full $\hat{\beta} = 0.338$, $p < 0.01$), with fit improving to $R^2 = 0.853$: falling from 0.454 in (1) to **0.338** in (8) but always positive and highly significant; Adjusted R^2 increases from 0.708 to 0.853. Commodities and the Dollar are strongly positive in (2)–(5) and become even larger when short rates and spreads are added (e.g., commodities 289 in (7)–(8); Dollar 273). VIX is strongly negative in (4) (−198.8) but turns positive and significant in (6)–(8) (29 to 173), indicating high-beta co-movement during risk-off episodes once other macros are held fixed. The 10Y rate flips from very positive in (5) (3056) to strongly negative with spreads (−1829 in (7)–(8)), while 1M is consistently very positive in (6)–(8) (1482→4211). The 10Y spread is strongly negative in (7)–(8) (−4983), and the 1Y spread turns strongly positive in (8) (5128), highlighting curve interactions. As in BTC, ZAR shows a robust positive ETH loading after rich macro controls—consistent with the FX channel in the model.

Cross-country synthesis and comparison to Bitcoin

Across the eight markets, the fully controlled ETH slopes are *uniformly positive and highly significant*: US (1.471), UK (0.231), Euro Area (0.036), Japan (0.034), China (0.012), Brazil (1.891), India (0.017), and South Africa (0.338). Fit metrics are high where global controls absorb large common variation (e.g., Japan $R^2 = 0.95$, India 0.95+), yet ETH’s positive loading survives throughout. This cross-market uniformity matches the model’s short-run parity logic: once we price the ETH leg in LCU, equity returns must load positively unless idiosyncratic wedges dominate.

Relative to BTC, the country ranking of magnitudes is broadly consistent (US, Brazil, and South Africa among the stronger; Euro Area, Japan, India, China smaller but precise), reinforcing that the FX-converted crypto leg acts as a common risk factor for local equities post-2020.

Together with our BTC results, the ETH evidence shows the “crypto–FX–equity” channel is not coin-specific: replacing Bitcoin with Ethereum preserves the sign and much of the cross-sectional pattern, strengthening the paper’s main argument that crypto is no longer a reliable diversifier against equities once measured through the FX lens in LCU.

Note on scale. Coefficients are estimated on log-difference units for both equities and ETH. Because crypto returns are much more volatile, slopes above those for BTC (e.g., US, BRL, ZAR) are not surprising. All magnitudes and inferences are internally consistent across the stepwise controls; for exposition we emphasize *sign*, *stability*, and *incremental R^2* , which are the model-relevant objects.

2.3 Litecoin (LTC)

Cross-Country Synthesis: LTC coefficients are overwhelmingly positive and significant across countries and specifications, with especially large elasticities in the U.S., Brazil, South Africa, and Japan (often in the multi-point range), and consistently positive but smaller slopes in Europe, India, and China. The only weak spot is an insignificant GBP coefficient in the most parsimonious column, which turns positive and significant once controls are included. Adding global and local rate controls boosts fit while leaving the LTC slope intact, mirroring the ETH pattern. Taken together, the LTC results provide further evidence against the hedge/diversifier narrative: Litecoin loads positively on local-currency equity returns nearly everywhere in the post-2020 sample.

2.3.1 USD

Table 17: Litecoin US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{LTC}_{USD})$	9.047*** (0.973)	5.232*** (0.818)	22.971*** (0.712)	22.460*** (0.669)	22.970*** (0.671)	23.978*** (0.672)
$\Delta \text{Commodity}$		90.246*** (2.575)	38.026*** (2.205)	30.951*** (2.102)	23.495*** (2.434)	34.256*** (2.710)
ΔDollar			371.593*** (7.811)	396.764*** (7.447)	366.761*** (8.946)	338.466*** (9.428)
ΔVIX				-74.551*** (3.851)	-58.146*** (4.711)	-43.646*** (4.949)
$\Delta 10\text{Y Yield}$					236.069*** (39.544)	-263.260*** (70.089)
$\Delta 1\text{M Yield}$						365.024*** (42.555)
Observations	2780	2780	2780	2780	2780	2780
Adj. R^2	0.030	0.327	0.629	0.673	0.677	0.685

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The $\Delta \log(\text{LTC}_{USD})$ coefficient is positive and highly significant in every column, rising from 9.05 in (1) to roughly 23–24 in (3)–(6); standard errors (0.67–0.97) are tight relative to the magnitudes. This implies an extremely high same-day equity sensitivity to LTC innovations—substantially larger than for BTC/ETH—consistent with a high-beta speculative channel rather than diversification. Commodities load strongly and positively (90 in (2), tapering to 23–34 by (5)–(6)) as other global factors are added. The Dollar index is large and positive in (3)–(6) (339–397), while VIX is strongly negative once included (–75 to –44), matching risk-on/risk-off intuition. Long rates turn from positive in (5) (236) to negative in (6) (–263) when the short end is added; the 1M rate itself is large and positive (365). Adjusted R^2 rises from 0.03 to 0.69 as controls absorb common shocks, but the LTC slope remains very large, supporting our core claim that crypto legs load positively on local equities.

2.3.2 UK

Table 18: Litecoin UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{GBP})$	2.099*** (0.396)	0.129 (0.334)	2.883*** (0.368)	2.269*** (0.277)	3.308*** (0.241)	3.128*** (0.243)	1.407*** (0.252)	2.212*** (0.252)
$\Delta \text{Commodity}$		26.212*** (0.741)	19.994*** (0.821)	15.082*** (0.626)	4.966*** (0.629)	3.479*** (0.706)	6.618*** (0.696)	13.080*** (0.833)
ΔDollar			43.126*** (2.831)	60.733*** (2.162)	20.363*** (2.266)	24.378*** (2.424)	10.166*** (2.447)	23.360*** (2.574)
ΔVIX				-52.859*** (1.143)	-30.751*** (1.212)	-32.752*** (1.285)	-39.002*** (1.274)	-39.186*** (1.236)
$\Delta 10Y \text{ Yield}$					317.412*** (10.175)	386.135*** (18.156)	185.212*** (20.827)	-10.093 (25.017)
$\Delta 1M \text{ Yield}$						-50.247*** (11.012)	-135.412*** (11.575)	92.171*** (20.535)
$\Delta 10Y \text{ Spread}$							270.252*** (15.682)	419.490*** (18.934)
$\Delta 3M \text{ Spread}$								-257.224*** (19.433)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.010	0.317	0.369	0.644	0.736	0.738	0.763	0.777

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's coefficient is positive and significant in all columns except (2) (0.129, ns), ranging from 2.10 in (1) to 3.31 in (5) and settling at 2.21 in (8) when yield-curve terms are included. Commodities are strongly positive early (26→15) and remain positive (3–13) after rates and spreads enter, indicating complementary macro channels rather than displacement. The Dollar is positive and significant throughout once included (10–61), VIX is consistently large and negative (–53 to –39), and long rates are strongly positive in (5)–(7) before turning insignificant in (8), while the 1M rate flips from negative in (6)–(7) to positive in (8) when curve shape is modeled. Spreads are sizable (10Y spread positive; 3M spread negative) in (7)–(8). Adjusted R^2 rises from 0.01 to 0.78; the LTC loading remains economically meaningful and robust.

2.3.3 Eurozone

Table 19: Litecoin Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{EUR})$	0.450*** (0.027)	0.288*** (0.022)	0.621*** (0.021)	0.590*** (0.018)	0.612*** (0.018)	0.648*** (0.018)	0.557*** (0.018)	0.472*** (0.019)
$\Delta \text{Commodity}$		2.216*** (0.057)	1.301*** (0.056)	1.044*** (0.048)	0.810*** (0.056)	1.131*** (0.061)	0.519*** (0.066)	0.564*** (0.065)
ΔDollar			6.226*** (0.190)	7.156*** (0.165)	6.240*** (0.199)	5.360*** (0.209)	4.292*** (0.204)	3.877*** (0.203)
ΔVIX				-2.815*** (0.088)	-2.310*** (0.107)	-1.869*** (0.111)	-2.731*** (0.114)	-2.611*** (0.112)
$\Delta 10\text{Y Yield}$					7.237*** (0.899)	-7.766*** (1.571)	-23.316*** (1.692)	-5.773** (2.260)
$\Delta 1\text{M Yield}$						10.996*** (0.954)	-1.265 (1.108)	-13.400*** (1.519)
$\Delta 10\text{Y Spread}$							40.119*** (2.122)	20.763*** (2.681)
$\Delta 3\text{M Spread}$								16.000*** (1.404)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.092	0.412	0.576	0.691	0.698	0.711	0.744	0.756

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's coefficient is uniformly positive and highly significant, moving from 0.45 in (1) to about 0.47–0.65 with controls (peaking 0.648 in (6)); standard errors are tiny (0.018–0.027). Commodities are positive across specifications (2.22→0.52–0.56), the Dollar is consistently positive (3.88–7.16), and VIX is reliably negative (−2.8 to −1.9). Long and short rates swap signs across columns—10Y turns positive in (5) then negative in (6)–(8), while 1M is positive in (6) and negative in (8)—and spreads are significantly positive in (7)–(8). Adjusted R^2 climbs from 0.09 to 0.76 as common macro factors are absorbed; the positive LTC loading persists throughout, reinforcing the paper's short-run parity prediction.

2.3.4 Japan

Table 20: Litecoin Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{JPY})$	0.906*** (0.042)	0.572*** (0.039)	1.089*** (0.030)	0.995*** (0.028)	1.015*** (0.026)	1.069*** (0.026)	0.453*** (0.017)	0.455*** (0.017)
$\Delta \text{Commodity}$		380.317*** (13.553)	116.291*** (11.354)	82.417*** (10.305)	-6.944 (11.264)	68.800*** (12.333)	198.990*** (7.259)	191.638*** (7.296)
ΔDollar			1727.200*** (35.569)	1867.669*** (32.482)	1463.367*** (39.705)	1251.562*** (41.703)	377.061*** (26.502)	377.884*** (26.311)
ΔVIX				-475.675*** (18.630)	-269.898*** (21.810)	-167.922*** (22.514)	-251.216*** (12.919)	-270.961*** (13.188)
$\Delta 10Y \text{ Yield}$					2955.984*** (181.089)	-562.522* (318.532)	-6402.989*** (197.830)	-6509.822*** (197.108)
$\Delta 1M \text{ Yield}$						2546.027*** (192.280)	2645.602*** (109.938)	2384.307*** (116.469)
$\Delta 10Y \text{ Spread}$							22774.155*** (301.344)	25711.267*** (546.113)
$\Delta 3M \text{ Spread}$								-6417.062*** (998.187)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.145	0.334	0.640	0.708	0.734	0.749	0.918	0.919

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's loading is positive and highly significant in all specifications: 0.57–1.09 through (6), then smaller (0.45) once powerful curve-shape terms enter in (7)–(8); Adjusted R^2 jumps to 0.92. Commodities swing from very large positive (380 in (2)) down to insignificant (−6.9 in (5)) and back to large positive with spreads (199–192), indicating macro channel reallocation. The Dollar is extremely large and positive (1727–1868 in (3)–(4)), declining as rates/spreads absorb variance (377 in (7)–(8)). VIX is strongly negative when added (−476), remaining negative thereafter. Long rates flip from huge positive (2956) to significantly negative with spreads (−6403; −6510), while the 1M rate is strongly positive (2546–2384). Spreads are massive (10Y spread 22.8k–25.7k; 3M spread −6417). Despite the heavy macro structure, the LTC slope stays positive and precise, consistent with our central claim.

2.3.5 China

Table 21: Litecoin China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{CNY})$	0.557*** (0.013)	0.526*** (0.013)	0.549*** (0.015)	0.560*** (0.014)	0.532*** (0.013)	0.540*** (0.014)	0.382*** (0.013)	0.378*** (0.013)
$\Delta \text{Commodity}$		4.069*** (0.255)	3.586*** (0.291)	4.038*** (0.290)	7.195*** (0.316)	7.800*** (0.358)	7.538*** (0.317)	8.210*** (0.322)
ΔDollar			3.430*** (0.999)	1.842* (0.997)	15.142*** (1.154)	13.525*** (1.237)	-14.466*** (1.498)	-9.188*** (1.588)
ΔVIX				5.142*** (0.537)	-1.959*** (0.620)	-1.155* (0.658)	-1.637*** (0.584)	-1.642*** (0.576)
$\Delta 10Y \text{ Yield}$					-101.694*** (5.176)	-129.334*** (9.299)	-115.827*** (8.263)	-135.402*** (8.428)
$\Delta 1M \text{ Yield}$						20.135*** (5.633)	2.349 (5.038)	4.235 (4.971)
$\Delta 10Y \text{ Spread}$							-309.176*** (11.271)	-442.839*** (18.485)
$\Delta 1Y \text{ Spread}$								174.696*** (19.309)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.387	0.439	0.441	0.459	0.525	0.527	0.627	0.638

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's coefficient is stable, positive, and precisely estimated: 0.53–0.56 through (6) and 0.38 in (7)–(8) after spreads are added; Adjusted R^2 increases from 0.39 to 0.64. Commodities are consistently positive and grow in magnitude (4.07→8.21), indicating a commodity-sensitive equity exposure. The Dollar is small positive in (3)–(4) but turns strongly positive in (5)–(6) and then flips negative with curve controls in (7)–(8) (−14.5; −9.19), consistent with RMB–USD dynamics once term structure is modeled. VIX moves from positive in (4) to negative in (5)–(8). Both long and short rates are mostly negative (10Y −102 to −135; 1M positive only in (6)), while spreads are strongly negative (10Y spread −309; −443) except for a positive 1Y spread in (8) (175). The persistent positive LTC loading supports the core paper claim even in a more frictional market.

2.3.6 Brazil

Table 22: Litecoin Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{BRL})$	20.463*** (1.113)	15.522*** (1.024)	24.824*** (0.983)	24.863*** (0.916)	24.771*** (0.942)	27.788*** (0.889)	33.121*** (0.915)	33.186*** (0.911)
$\Delta \text{Commodity}$		443.436*** (17.658)	227.938*** (17.819)	161.631*** (16.905)	166.140*** (20.148)	365.056*** (21.084)	490.322*** (21.676)	347.120*** (34.243)
ΔDollar			1532.518*** (58.436)	1776.331*** (55.696)	1793.285*** (69.285)	1212.441*** (70.363)	1511.016*** (69.953)	1410.594*** (72.058)
ΔVIX				-644.505*** (31.261)	-653.869*** (38.669)	-381.937*** (38.331)	-280.440*** (37.257)	-274.341*** (37.088)
$\Delta 10Y \text{ Yield}$					-135.790 (329.977)	-9287.185*** (538.930)	-2902.982*** (654.199)	-2083.034*** (668.500)
$\Delta 1M \text{ Yield}$						6769.296*** (327.530)	4833.398*** (336.498)	4160.577*** (357.374)
$\Delta 10Y \text{ Spread}$							-3385.254*** (213.193)	-4542.492*** (301.990)
$\Delta 1Y \text{ Spread}$								1270.677*** (236.014)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.108	0.273	0.417	0.494	0.494	0.562	0.598	0.602

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's slope is extremely large and highly significant across all models: 15.5–33.2, with the peak in the fully controlled specifications (7)–(8). Commodities and the Dollar both load very strongly and positively (commodities 443→347–490; Dollar 1533→1776–1793, easing to 1212–1511 as short rates/spreads enter). VIX is sharply negative (–645 to –274). The 10Y yield becomes strongly negative once the short end is included (–9287 in (6); –2903/–2083 with spreads), while the 1M rate is very large and positive (6769→4161). Spreads are economically large and significant (10Y spread negative; 1Y spread positive). Adjusted R^2 rises to 0.60. The persistent and very large LTC loading further undermines any hedge/diversifier narrative.

2.3.7 India

Table 23: Litecoin India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{\text{INR}})$	0.329*** (0.026)	0.139*** (0.020)	0.484*** (0.017)	0.459*** (0.015)	0.490*** (0.014)	0.529*** (0.014)	0.110*** (0.011)	0.108*** (0.011)
$\Delta \text{Commodity}$		200.641*** (4.687)	106.700*** (3.937)	91.969*** (3.598)	56.883*** (3.980)	95.162*** (4.213)	114.511*** (2.786)	87.353*** (3.094)
ΔDollar			636.853*** (13.023)	690.939*** (11.937)	546.629*** (14.104)	440.782*** (14.287)	92.661*** (11.008)	100.988*** (10.487)
ΔVIX				-165.675*** (6.557)	-88.331*** (7.681)	-37.424*** (7.657)	-47.794*** (5.032)	-38.817*** (4.817)
$\Delta 10\text{Y Yield}$					1107.898*** (64.298)	-648.222*** (108.119)	2107.082*** (84.364)	2506.925*** (83.650)
$\Delta 1\text{M Yield}$						1280.568*** (65.448)	597.253*** (44.442)	919.890*** (46.346)
$\Delta 10\text{Y Spread}$							-4813.528*** (79.592)	-2142.724*** (174.212)
$\Delta 2\text{Y Spread}$								-1976.238*** (116.085)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.055	0.431	0.694	0.751	0.775	0.802	0.915	0.923

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

LTC's coefficient is positive and significant throughout, increasing with initial controls (0.33→0.53 by (6)) and then shrinking once yield-curve terms are added (0.11 in (7)–(8)). Commodities remain strongly positive (201→87), Dollar effects are large and positive (637→691, moderating to 93–101 after curve controls), and VIX is strongly negative once included (−166→−39). The 10Y rate flips sign (positive in (5), negative in (6), then strongly positive again with spreads in (7)–(8)), while the 1M rate is consistently large and positive in (6)–(8) (1281→920). Spreads are very large (10Y spread −4814; −2143; 2Y spread −1976). Fit improves dramatically (Adj. R^2 0.06→0.92). The persistent positive LTC loading supports the parity-based linkage even with dense macro controls.

2.3.8 South Africa

Table 24: Litecoin South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{LTC}_{ZAR})$	6.720*** (0.321)	4.066*** (0.254)	6.898*** (0.223)	6.559*** (0.207)	7.070*** (0.197)	7.266*** (0.194)	8.076*** (0.182)	8.076*** (0.182)
$\Delta \text{Commodity}$		495.691*** (11.346)	316.299*** (10.547)	282.210*** (9.907)	176.521*** (10.897)	237.422*** (12.172)	416.218*** (13.660)	416.218*** (13.660)
ΔDollar			1238.149*** (33.945)	1366.786*** (32.039)	928.710*** (38.140)	758.457*** (40.802)	1006.422*** (39.012)	1006.422*** (39.012)
ΔVIX				-388.867*** (18.220)	-155.672*** (21.182)	-78.926*** (22.037)	102.106*** (21.739)	102.106*** (21.739)
$\Delta 10Y \text{ Yield}$					3329.619*** (177.222)	634.311** (310.828)	-2661.961*** (320.019)	-2661.961*** (320.019)
$\Delta 1M \text{ Yield}$						1958.341*** (187.208)	5116.504*** (220.902)	5116.504*** (220.902)
$\Delta 10Y \text{ Spread}$							-4826.525*** (212.079)	-4826.525*** (212.079)
$\Delta 1Y \text{ Spread}$								-7750.959*** (505.894)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.136	0.488	0.654	0.702	0.736	0.746	0.786	0.786

Note: Dependent variable shown in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The LTC coefficient is large, positive, and highly significant in all columns, rising from 6.72 in (1) to about 8.08 in (7)–(8). Commodities and the Dollar both load very strongly and positively (commodities 496→416; Dollar 1238→1367, easing to 758–1006 with short rates and spreads). VIX is sharply negative when first included (–389), becomes less negative as rates enter (–156; –79), and turns positive with curve controls (102), consistent with high-beta behavior after holding other macros fixed. The 10Y yield is very positive in (5), becomes modestly positive in (6), and flips strongly negative with curve terms (–2662), while the 1M rate is very positive in (6)–(8) (1958→5117). Spreads are economically large (10Y spread –4827; 1Y spread –7751). Model fit improves to Adj. $R^2 \approx 0.79$. The positive LTC loading persists across all specifications, reinforcing the main thesis.

2.4 Tether

USDT results are heterogeneous across countries—robustly negative in GBP/EUR/CNY, robustly positive in JPY/BRL/INR/ZAR, and indistinguishable from zero in the US once controls enter—consistent with a USD-pegged crypto primarily transmitting the USD/FX leg rather than independent “crypto risk.”

2.4.1 US

Table 25: Tether US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{USDT}_{USD})$	-18731.357** (8165.569)	-1061.363 (6731.488)	6170.717 (5578.994)	4475.703 (5366.586)	4842.761 (5386.921)	2256.227 (5401.686)
$\Delta \text{Commodity}$		98.140*** (2.604)	63.355*** (2.354)	56.146*** (2.311)	55.082*** (2.674)	62.107*** (3.101)
ΔDollar			258.559*** (7.017)	292.270*** (7.088)	287.547*** (9.266)	265.635*** (10.477)
ΔVIX				-68.300*** (4.394)	-65.772*** (5.432)	-58.537*** (5.656)
$\Delta 10\text{Y Yield}$					36.161 (45.684)	-263.359*** (81.504)
$\Delta 1\text{M Yield}$						216.752*** (48.915)
Observations	2968	2968	2968	2968	2968	2968
Adj. R^2	0.001	0.325	0.537	0.571	0.571	0.574

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical interpretation—US. Column (1) shows a large negative and statistically significant USDT loading in USD (-18731.357; $p < 0.05$) that becomes statistically indistinguishable from zero once controls are added (cols. 2–6). The stepwise entry of global factors behaves as expected—commodities and the dollar index are strongly positive, VIX is strongly negative, and the short rate enters positively while the 10Y yield turns significantly negative only in col. (6). Adjusted R^2 rises from 0.001 to 0.574 by col. (6), indicating controls explain the equity variation; the USDT slope itself is not robustly different from zero once common risk is absorbed, consistent with a USD-pegged asset carrying little independent “crypto risk” in USD terms.

2.4.2 UK

Table 26: Tether UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{GBP})$	-613.629*	-3266.831***	-9061.857***	-7570.478***	-6848.709***	-6734.439***	-5640.525***	-7745.609***
	(350.477)	(293.818)	(355.235)	(258.947)	(224.869)	(226.650)	(261.684)	(258.677)
$\Delta \text{Commodity}$		27.556***	21.684***	16.552***	8.394***	7.295***	8.101***	19.185***
		(0.712)	(0.690)	(0.510)	(0.510)	(0.593)	(0.595)	(0.735)
ΔDollar			67.620***	85.082***	45.327***	48.148***	39.164***	67.630***
			(2.728)	(2.005)	(2.137)	(2.273)	(2.506)	(2.631)
ΔVIX				-50.539***	-31.536***	-32.706***	-34.737***	-34.801***
				(0.975)	(1.034)	(1.082)	(1.099)	(1.014)
$\Delta 10Y \text{ Yield}$					276.813***	323.586***	235.265***	-57.265***
					(8.720)	(15.638)	(18.921)	(21.697)
$\Delta 1M \text{ Yield}$						-33.737***	-67.592***	294.816***
						(9.372)	(10.168)	(18.508)
$\Delta 10Y \text{ Spread}$							120.277***	317.788***
							(14.836)	(16.219)
$\Delta 3M \text{ Spread}$								-397.570***
								(17.500)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.001	0.336	0.450	0.711	0.784	0.785	0.790	0.821

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical interpretation—UK. The USDT–equity slope in GBP is unambiguously *negative and highly significant* across all controlled specifications (cols. 2–8), ranging from -.857 to 3266.831. Adding the dollar index, VIX, and local yield controls raises R^2 from 0.001 to 0.821 and leaves the USDT slope negative and precise. Control signs are conventional: Dollar and 10Y level positive (until term-structure saturation flips the 10Y in col. 8), VIX negative, and a strongly negative 3M spread in the final column. The pattern is consistent with USDT in GBP acting primarily as a *USD factor proxy*: when USD strengthens (and USDT_{GBP} rises), UK equities in LCU tend to fall.

2.4.3 Eurozone

Table 27: Tether Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{EUR})$	356.249*** (22.832)	-70.519*** (21.556)	-1161.499*** (25.990)	-1173.149*** (20.622)	-1242.752*** (20.845)	-1269.439*** (20.179)	-1291.328*** (22.970)	-1184.047*** (22.871)
$\Delta \text{Commodity}$		2.528*** (0.065)	2.640*** (0.047)	2.350*** (0.038)	2.657*** (0.044)	3.026*** (0.049)	3.101*** (0.062)	2.905*** (0.061)
ΔDollar			11.499*** (0.220)	12.981*** (0.178)	14.623*** (0.217)	13.753*** (0.217)	13.969*** (0.243)	12.805*** (0.242)
ΔVIX				-2.835*** (0.068)	-3.462*** (0.083)	-3.116*** (0.083)	-3.074*** (0.086)	-2.850*** (0.083)
$\Delta 10Y \text{ Yield}$					-8.932*** (0.711)	-23.794*** (1.206)	-22.439*** (1.384)	-2.598 (1.776)
$\Delta 1M \text{ Yield}$						10.647*** (0.711)	11.579*** (0.851)	-2.387** (1.165)
$\Delta 10Y \text{ Spread}$							-3.502** (1.758)	-22.413*** (2.025)
$\Delta 3M \text{ Spread}$								17.798*** (1.063)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.076	0.389	0.682	0.800	0.810	0.823	0.824	0.839

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical interpretation—Euro Area. USDT_{EUR} loads positively in the bivariate regression (356; col. 1) but turns negative and highly significant once common factors are included (1161.499 to 1291.328, cols. 3–8). Dollar and commodities are positive; VIX is negative; 10Y and 1M rates capture term-structure dynamics with expected signs, and spreads improve fit further. Adjusted R^2 climbs from 0.076 to 0.839. The negative USDT slope in LCU is consistent with USDT acting as a USD-strength proxy against the euro: when USDT_{EUR} rises (USD up), euro-area equities in LCU tend to fall.

2.4.4 Japan

Table 28: Tether Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{JPY})$	625.918*** (6.574)	649.545*** (8.695)	772.846*** (13.626)	795.088*** (16.976)	1236.761*** (21.181)	1288.243*** (22.063)	629.859*** (17.518)	657.466*** (18.000)
$\Delta \text{Commodity}$		-42.887*** (10.367)	-76.874*** (10.557)	-82.741*** (10.884)	-75.817*** (9.580)	-126.057*** (11.590)	98.911*** (8.251)	96.433*** (8.212)
ΔDollar			-462.478*** (39.927)	-535.349*** (51.914)	-893.866*** (47.278)	-855.532*** (47.111)	-512.996*** (30.852)	-547.410*** (31.189)
ΔVIX				44.386** (20.229)	-71.535*** (18.231)	-102.742*** (18.527)	-187.310*** (12.024)	-167.902*** (12.373)
$\Delta 10Y \text{ Yield}$					-6168.637*** (209.738)	-4817.551*** (274.180)	-8157.339*** (184.303)	-8043.250*** (184.167)
$\Delta 1M \text{ Yield}$						-1247.150*** (165.128)	989.926*** (112.033)	1132.151*** (113.811)
$\Delta 10Y \text{ Spread}$							20068.070*** (311.317)	17084.306*** (581.612)
$\Delta 3M \text{ Spread}$								5870.264*** (968.853)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.753	0.755	0.765	0.766	0.818	0.822	0.926	0.927

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Empirical interpretation—Japan. USDT_{JPY} displays a *large, positive, and highly significant* loading throughout (+625.918 to +1,288.243). Dollar enters strongly negative (as USD appreciation coincides with JPY depreciation), VIX switches from slightly positive to negative once rates are included, and yield terms show the usual BOJ-driven curve effects (big negative 10Y level; positive spreads raising fit). Adjusted R^2 is very high (0.753→0.927), and the USDT slope remains robustly positive—consistent with yen-denominated USDT acting like a USD risk factor that comoves with Japanese equities in LCU over this sample.

2.4.5 China

Table 29: Tether China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{CNY})$	-315.822*** (14.991)	-331.780*** (14.165)	-425.068*** (27.472)	-527.164*** (32.846)	-325.586*** (36.058)	-313.540*** (39.758)	-341.235*** (33.550)	-450.333*** (32.308)
$\Delta \text{Commodity}$		5.564*** (0.290)	4.697*** (0.363)	3.599*** (0.411)	7.687*** (0.522)	7.643*** (0.526)	5.545*** (0.448)	6.409*** (0.426)
ΔDollar			7.241*** (1.829)	15.101*** (2.297)	13.446*** (2.246)	13.357*** (2.250)	-18.607*** (2.110)	-2.851 (2.172)
ΔVIX				-4.106*** (0.733)	-7.515*** (0.768)	-7.619*** (0.781)	-7.570*** (0.659)	-6.832*** (0.625)
$\Delta 10Y \text{ Yield}$					-84.639*** (6.933)	-78.503*** (10.991)	-56.152*** (9.294)	-115.337*** (9.363)
$\Delta 1M \text{ Yield}$						-5.240 (7.284)	-21.909*** (6.163)	0.530 (5.960)
$\Delta 10Y \text{ Spread}$							-397.114*** (11.461)	-688.734*** (19.129)
$\Delta 1Y \text{ Spread}$								358.592*** (19.371)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.130	0.226	0.230	0.237	0.274	0.274	0.483	0.537

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Empirical interpretation—China. USDT_{CNY} coefficients are *negative and highly significant* in all columns (-315.822 to -527.164 initially; 341.540 to -450.333 with full controls). Dollar turns from positive to (insignificant) after spreads, VIX is consistently negative, and policy-rate terms carry the expected signs for a managed FX/capital-control environment. Fit improves with curve controls (Adj. R^2 from 0.13 to 0.537). The negative USDT slope aligns with USD strength weighing on CNY-denominated equities.

2.4.6 Brazil

Table 30: Tether Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{BRL})$	15825.660*** (301.071)	13817.545*** (300.873)	13230.192*** (377.486)	16441.818*** (346.958)	18569.890*** (377.240)	17924.965*** (356.334)	22800.778*** (326.814)	23593.125*** (320.815)
$\Delta \text{Commodity}$		313.005*** (15.870)	302.237*** (16.398)	195.173*** (14.778)	62.058*** (17.808)	254.402*** (19.395)	428.866*** (16.896)	139.137*** (25.958)
ΔDollar			157.081** (61.069)	277.316*** (53.571)	-422.837*** (75.926)	-903.864*** (75.484)	-1053.266*** (63.124)	-1319.908*** (63.803)
ΔVIX				-878.415*** (29.117)	-656.878*** (33.300)	-466.417*** (32.783)	-398.615*** (27.421)	-380.135*** (26.548)
$\Delta 10Y \text{ Yield}$					3962.418*** (312.140)	-3922.545*** (496.808)	10118.595*** (569.566)	12344.328*** (572.167)
$\Delta 1M \text{ Yield}$						5569.575*** (283.090)	967.837*** (268.676)	-499.112* (279.168)
$\Delta 10Y \text{ Spread}$							-6058.501*** (168.523)	-8575.319*** (239.293)
$\Delta 1Y \text{ Spread}$								2551.697*** (177.651)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.482	0.542	0.543	0.650	0.668	0.706	0.796	0.809

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Empirical interpretation—Brazil. USDT_{BRL} loads *strongly positive* throughout (13817.545 to 23593.125), with precision and magnitude increasing as the specification enriches (Adj. R^2 0.482→0.809). The dollar coefficient flips from positive in early columns to large negative once rates and spreads are included, reflecting BRL’s joint exposure to USD and local curve moves. VIX is strongly negative; term-structure terms are economically large. The robust positive USDT slope suggests BRL-denominated stablecoin changes comove with Brazilian equities in LCU over this period.

2.4.7 India

Table 31: Tether India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{\text{INR}})$	677.877*** (6.362)	581.024*** (6.949)	786.831*** (11.994)	748.561*** (10.886)	737.366*** (10.859)	789.554*** (11.911)	433.316*** (13.436)	308.856*** (14.914)
$\Delta \text{Commodity}$		76.026*** (3.030)	65.352*** (2.885)	56.849*** (2.615)	46.851*** (2.869)	24.913*** (3.587)	72.296*** (3.178)	61.328*** (3.114)
ΔDollar			-289.533*** (14.174)	-192.047*** (13.272)	-229.811*** (13.941)	-224.481*** (13.728)	-125.805*** (11.504)	-71.059*** (11.502)
ΔVIX				-120.273*** (4.547)	-94.707*** (5.504)	-112.215*** (5.696)	-69.451*** (4.784)	-56.594*** (4.645)
$\Delta 10\text{Y Yield}$					375.508*** (46.572)	1086.344*** (85.071)	1878.142*** (72.477)	2405.222*** (76.392)
$\Delta 1\text{M Yield}$						-538.510*** (54.298)	-16.543 (46.384)	424.933*** (51.854)
$\Delta 10\text{Y Spread}$							-3147.019*** (81.862)	-1312.181*** (136.145)
$\Delta 2\text{Y Spread}$								-1818.598*** (110.349)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.793	0.829	0.850	0.879	0.881	0.885	0.923	0.930

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical interpretation—India. USDT_{INR} coefficients are *positive and highly significant* in all columns (581.024 to 789.554 initially; 308.856 in the final saturated model). The dollar coefficient is negative—as USD strength typically coincides with INR weakness and weaker local equities—while VIX is also negative. Yield levels and spreads substantially raise fit (Adj. R^2 reaches 0.93). The USDT slope remains positive after controlling for these macro channels, indicating a persistent same-day comovement in LCU.

2.4.8 South Africa

Table 32: Tether South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{USDT}_{ZAR})$	3734.927*** (87.317)	2623.885*** (73.717)	3255.047*** (144.581)	3123.029*** (134.840)	3082.945*** (131.619)	3910.210*** (162.578)	7592.822*** (160.579)	7592.822*** (160.579)
$\Delta \text{Commodity}$		424.683*** (10.015)	435.229*** (10.189)	394.016*** (9.688)	326.294*** (10.954)	259.794*** (13.367)	520.176*** (12.619)	520.176*** (12.619)
ΔDollar			-303.240*** (59.843)	-62.025 (56.895)	-346.499*** (60.189)	-427.152*** (60.236)	-976.739*** (50.548)	-976.739*** (50.548)
ΔVIX				-392.460*** (18.461)	-233.573*** (22.206)	-301.948*** (23.378)	-112.937*** (19.460)	-112.937*** (19.460)
$\Delta 10Y \text{ Yield}$					2281.954*** (186.477)	5127.118*** (382.736)	3680.866*** (311.089)	3680.866*** (311.089)
$\Delta 1M \text{ Yield}$						-2093.250*** (246.798)	263.671 (207.840)	263.671 (207.840)
$\Delta 10Y \text{ Spread}$							-9468.583*** (237.910)	-9468.583*** (237.910)
$\Delta 1Y \text{ Spread}$								12423.757*** (580.066)
Observations	2968	2968	2968	2968	2968	2968	2968	2968
Adj. R^2	0.381	0.615	0.618	0.668	0.684	0.692	0.799	0.799

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Empirical interpretation—South Africa. USDT_{ZAR} coefficients are *positive and precisely estimated* in every column (2623.885 to +7592.822), with fit improving from 0.381 to 0.799 as macro controls accumulate. Dollar turns strongly negative after rates are added (cols. 5–8), VIX is negative, and yield terms are economically large—especially the 10Y spread in the final model. The positive USDT slope indicates same-day comovement of ZAR-denominated stablecoin with JSE equities in LCU.

2.5 Ripple

Ripple (XRP) Cross-Country synthesis. Except for the spread-heavy Japan specifications (cols. 7–8), XRP loadings are uniformly positive and highly significant across countries and control sets; adjusted R^2 rises sharply as controls enter while the XRP slope survives. This pattern is consistent with our main claim: in the post-2020 period crypto behaves as a pro-equity factor in local currency rather than a diversifier or hedge.

2.5.1 US

Table 33: Ripple US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{XRP}_{USD})$	2632.913*** (50.888)	2192.548*** (45.722)	1935.933*** (40.378)	1838.690*** (39.211)	1892.682*** (39.565)	2301.107*** (38.534)
$\Delta \text{Commodity}$		63.465*** (1.995)	45.017*** (1.826)	40.413*** (1.775)	47.426*** (2.009)	78.192*** (2.135)
ΔDollar			181.228*** (5.899)	204.520*** (5.849)	238.122*** (7.429)	145.710*** (7.491)
ΔVIX				-53.989*** (3.451)	-70.828*** (4.138)	-33.356*** (3.955)
$\Delta 10\text{Y Yield}$					-254.952*** (35.268)	-1669.975*** (61.890)
$\Delta 1\text{M Yield}$						969.472*** (36.502)
Observations	2780	2780	2780	2780	2780	2780
Adj. R^2	0.491	0.626	0.721	0.744	0.748	0.799

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The XRP_{USD} slope is large, positive, and highly significant across all six columns (from 2632.9 in the bivariate to 2301.1 with the full control set). Magnitudes drift down as global factors enter (cols. 2–4), then partially rebound once rate levels are included and the 1M yield is added (cols. 5–6), indicating XRP’s equity comovement is not subsumed by the Dollar, VIX, or commodities. Signs on controls follow intuition: the VIX is strongly negative, Dollar strength loads positively in LCU terms, and the long rate becomes sharply negative with short-rate controls (col. 6), consistent with discount-rate channels. Adjusted R^2 rises from 0.49 to 0.80, while the XRP coefficient remains robust—supporting our short-run positive co-movement thesis for the U.S.

2.5.2 UK

Table 34: Ripple UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{GBP})$	910.031*** (20.383)	732.977*** (18.639)	703.825*** (19.112)	586.194*** (13.878)	520.448*** (12.478)	574.584*** (13.451)	627.849*** (17.992)	613.496*** (17.670)
$\Delta \text{Commodity}$		18.621*** (0.617)	17.359*** (0.646)	13.386*** (0.469)	6.975*** (0.473)	9.942*** (0.555)	9.829*** (0.554)	14.665*** (0.700)
ΔDollar			12.990*** (2.110)	33.901*** (1.565)	3.532** (1.756)	-5.664*** (1.965)	-5.042** (1.963)	2.422 (2.041)
ΔVIX				-46.936*** (0.914)	-31.585*** (0.976)	-27.942*** (1.028)	-25.982*** (1.116)	-27.163*** (1.098)
$\Delta 10Y \text{ Yield}$					232.829*** (8.313)	96.189*** (16.139)	123.482*** (17.220)	-16.144 (21.174)
$\Delta 1M \text{ Yield}$						93.509*** (9.524)	128.252*** (12.303)	268.569*** (17.627)
$\Delta 10Y \text{ Spread}$							-71.878*** (16.194)	51.515*** (19.482)
$\Delta 3M \text{ Spread}$								-174.550*** (16.004)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.418	0.561	0.567	0.778	0.827	0.833	0.834	0.840

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

XRP_{GBP} coefficients are uniformly positive and highly significant across all eight columns, falling from 910.031 (baseline) to a still-large 613.496 with the richest controls. Fit improves markedly (Adj. R^2 0.42→0.84). Dollar sensitivity is positive early on but turns mildly negative once short rates are included (col. 6–7), a known interaction in LCU settings; VIX is strongly negative throughout. The stability of the XRP slope in the presence of Dollar/VIX/rates confirms persistent short-run co-movement with U.K. equities.

2.5.3 Eurozone

Table 35: Ripple Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{EUR})$	67.233*** (1.354)	53.460*** (1.173)	49.713*** (1.182)	44.575*** (1.047)	45.434*** (1.061)	57.906*** (1.017)	52.735*** (1.056)	53.991*** (0.883)
$\Delta \text{Commodity}$		1.688*** (0.046)	1.506*** (0.047)	1.305*** (0.042)	1.408*** (0.048)	2.201*** (0.049)	1.723*** (0.059)	1.783*** (0.050)
ΔDollar			1.903*** (0.154)	2.975*** (0.140)	3.464*** (0.176)	0.963*** (0.176)	0.685*** (0.171)	0.316** (0.144)
ΔVIX				-2.388*** (0.081)	-2.635*** (0.098)	-1.649*** (0.092)	-2.210*** (0.098)	-1.634*** (0.083)
$\Delta 10Y \text{ Yield}$					-3.746*** (0.833)	-40.396*** (1.437)	-47.226*** (1.480)	-10.881*** (1.621)
$\Delta 1M \text{ Yield}$						25.122*** (0.850)	16.169*** (1.054)	-4.522*** (1.064)
$\Delta 10Y \text{ Spread}$							25.148*** (1.852)	-20.412*** (2.030)
$\Delta 3M \text{ Spread}$								32.340*** (0.933)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.470	0.643	0.661	0.742	0.743	0.805	0.817	0.872

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

XRP_{EUR} coefficients remain positive and tightly estimated across all eight columns (67.2 in the baseline, settling near 44–58 with controls). Adding Dollar/VIX and then rate levels/spreads steadily raises fit (Adj. R^2 up to 0.87) without eliminating the XRP slope. Dollar enters positively but attenuates once short rates and term structure are included, while VIX is consistently negative. The persistence of a positive XRP loading confirms short-run pro-equity behavior in the Euro Area.

2.5.4 Japan

Table 36: Ripple Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{JPY})$	86.365*** (1.741)	72.349*** (1.641)	60.049*** (1.556)	53.681*** (1.448)	51.706*** (1.474)	65.679*** (1.457)	-10.325*** (1.591)	-8.089*** (1.755)
$\Delta \text{Commodity}$		288.789*** (10.845)	210.299*** (10.253)	173.817*** (9.500)	141.432*** (10.799)	298.671*** (11.751)	252.802*** (7.791)	250.797*** (7.809)
ΔDollar			867.698*** (34.221)	1075.785*** (32.517)	926.936*** (40.319)	435.911*** (41.826)	157.137*** (27.986)	155.352*** (27.953)
ΔVIX				-436.231*** (18.635)	-359.681*** (22.286)	-170.541*** (21.686)	-344.469*** (14.600)	-350.649*** (14.725)
$\Delta 10Y \text{ Yield}$					1178.517*** (191.043)	-5944.213*** (340.694)	-6616.328*** (225.078)	-6750.416*** (229.185)
$\Delta 1M \text{ Yield}$						4845.888*** (199.478)	1788.063*** (141.150)	1738.200*** (141.931)
$\Delta 10Y \text{ Spread}$							29188.068*** (486.652)	30340.769*** (620.269)
$\Delta 3M \text{ Spread}$								-3696.303*** (1236.033)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.469	0.577	0.657	0.713	0.717	0.767	0.898	0.899

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

For Japan, XRP_{JPY} is positive and highly significant through columns (1)–(6), ranging from 86.4 to 65.7 as controls accumulate. Once the full set of rate *spreads* is introduced (cols. 7–8), the coefficient turns small and negative (about 10 to 8 bps, significant). This flip coincides with extremely large and precisely estimated term-structure loadings, suggesting multicollinearity/over-control when both levels and spreads (plus Dollar/VIX) jointly absorb equity variation. Fit nonetheless jumps to 0.90. We flag this as a sensitivity: the pre-spread specifications (cols. 1–6) support the main claim (positive co-movement), whereas the spread-heavy variants warrant robustness checks (e.g., excluding one of the spread blocks or using rolling samples).

2.5.5 China

Table 37: Ripple China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{CNY})$	25.875*** (0.949)	22.532*** (0.972)	27.715*** (0.926)	29.526*** (0.919)	34.921*** (0.792)	40.294*** (0.824)	34.047*** (1.363)	33.899*** (1.470)
$\Delta \text{Commodity}$		3.533*** (0.306)	5.649*** (0.298)	6.219*** (0.295)	10.870*** (0.286)	13.827*** (0.326)	12.989*** (0.356)	12.994*** (0.356)
ΔDollar			-21.597*** (0.978)	-24.588*** (0.991)	-2.785*** (1.060)	-11.721*** (1.148)	-17.471*** (1.519)	-17.355*** (1.579)
ΔVIX				6.650*** (0.578)	-4.392*** (0.589)	-0.850 (0.602)	-1.325** (0.604)	-1.338** (0.606)
$\Delta 10Y \text{ Yield}$					-168.443*** (5.037)	-302.325*** (9.441)	-269.430*** (11.000)	-269.325*** (11.009)
$\Delta 1M \text{ Yield}$						91.403*** (5.547)	72.422*** (6.432)	72.064*** (6.569)
$\Delta 10Y \text{ Spread}$							-99.782*** (17.396)	-105.723*** (28.086)
$\Delta 1Y \text{ Spread}$								5.847 (21.700)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.211	0.247	0.359	0.388	0.564	0.602	0.607	0.607

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

XRP_{CNY} is positive and highly significant in every column, rising from 25.9 (baseline) to 40.3 once rate levels are added (col. 6), and remaining around 34.0 with spreads (cols. 7–8). Dollar coefficients are negative (LCU mechanics), VIX turns from positive (without rates) to negative with macro controls, and long rates are strongly negative—consistent with discount-rate effects in China’s market structure. Model fit climbs from 0.21 to 0.61, while the XRP slope persists, supporting our claim.

2.5.6 Brazil

Table 38: Ripple Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{BRL})$	2276.903*** (66.934)	1858.132*** (65.296)	1661.040*** (66.604)	1508.719*** (64.276)	1595.121*** (63.731)	2320.431*** (59.905)	3319.561*** (60.246)	3539.916*** (61.732)
$\Delta \text{Commodity}$		350.200*** (16.661)	292.297*** (17.205)	245.212*** (16.679)	337.581*** (18.787)	656.608*** (19.268)	996.836*** (19.696)	1294.738*** (31.829)
ΔDollar			607.830*** (56.846)	844.617*** (56.170)	1279.282*** (70.110)	244.899*** (69.297)	383.601*** (59.545)	524.387*** (59.350)
ΔVIX				-532.667*** (32.466)	-752.950*** (38.699)	-353.625*** (35.870)	-138.491*** (31.480)	-118.748*** (30.776)
$\Delta 10Y \text{ Yield}$					-3283.686*** (326.693)	-17833.353*** (551.300)	-9857.538*** (535.310)	-11492.091*** (540.776)
$\Delta 1M \text{ Yield}$						10116.125*** (329.272)	8156.466*** (288.869)	9651.891*** (309.404)
$\Delta 10Y \text{ Spread}$							-6182.075*** (195.102)	-4386.979*** (244.204)
$\Delta 1Y \text{ Spread}$								-2392.505*** (203.716)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.294	0.391	0.414	0.466	0.485	0.615	0.718	0.731

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In Brazil, XRP_{BRL} is positive and highly significant in every specification, starting near 2276.9 and rising to 3540 with the richest control set as commodities, Dollar, VIX, and interest-rate blocks are added. Fit improves substantially (Adj. R^2 0.29→0.73). Large negative long-rate and spread loadings reflect strong discount-rate sensitivity in IBOV; nevertheless, the XRP slope strengthens with controls, reinforcing the equity–crypto linkage in a high-beta LCU environment.

2.5.7 India

Table 39: Ripple India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{XRP}_{INR})$	52.238*** (1.245)	39.124*** (1.034)	31.434*** (0.922)	28.285*** (0.863)	27.195*** (0.873)	36.886*** (0.814)	4.810*** (0.833)	2.894*** (0.805)
$\Delta \text{Commodity}$		161.410*** (3.942)	126.078*** (3.568)	113.796*** (3.340)	101.904*** (3.792)	167.689*** (3.896)	127.021*** (2.848)	98.307*** (3.249)
ΔDollar			368.453*** (11.731)	433.203*** (11.217)	377.387*** (14.096)	175.015*** (13.788)	45.366*** (10.011)	54.693*** (9.588)
ΔVIX				-143.524*** (6.517)	-115.162*** (7.819)	-35.701*** (7.209)	-50.945*** (5.084)	-43.759*** (4.881)
$\Delta 10Y \text{ Yield}$					430.819*** (66.701)	-2559.121*** (112.227)	1924.201*** (115.636)	2501.137*** (116.145)
$\Delta 1M \text{ Yield}$						2049.890*** (66.084)	660.682*** (53.382)	906.572*** (53.243)
$\Delta 10Y \text{ Spread}$							-4903.972*** (92.340)	-2438.731*** (175.982)
$\Delta 2Y \text{ Spread}$								-1929.917*** (119.182)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.387	0.618	0.718	0.760	0.763	0.824	0.913	0.920

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

XRP_{INR} is positive and precisely estimated in all columns; magnitudes attenuate from 52 (baseline) to 27 with global controls (cols. 4–5), bounce with rate levels (col. 6), and compress again to small but significant values (4.8/2.9) when spreads are added (cols. 7–8). The sharp $Adj.R^2$ increase (0.39→0.92) shows controls explain substantial common variation, yet the XRP slope remains strictly positive—consistent with our thesis for India at the daily horizon.

2.5.8 South Africa

Table 40: Ripple South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(XRP_{ZAR})$	758.148*** (12.944)	605.323*** (10.010)	556.890*** (10.097)	524.607*** (9.602)	513.820*** (9.743)	631.738*** (8.655)	632.121*** (8.950)	632.121*** (8.950)
$\Delta \text{Commodity}$		388.108*** (7.965)	350.031*** (8.025)	324.791*** (7.628)	301.083*** (8.690)	470.546*** (8.543)	469.484*** (10.620)	469.484*** (10.620)
ΔDollar			416.185*** (26.798)	553.332*** (26.060)	443.935*** (32.465)	-86.073*** (30.460)	-87.811*** (32.166)	-87.811*** (32.166)
ΔVIX				-293.596*** (14.928)	-237.360*** (17.928)	-34.090** (15.789)	-34.953** (16.602)	-34.953** (16.602)
$\Delta 10Y \text{ Yield}$					856.515*** (153.060)	-6781.797*** (245.258)	-6767.001*** (260.561)	-6767.001*** (260.561)
$\Delta 1M \text{ Yield}$						5236.424*** (144.236)	5220.738*** (171.722)	5220.738*** (171.722)
$\Delta 10Y \text{ Spread}$							28.302 (168.063)	28.302 (168.063)
$\Delta 1Y \text{ Spread}$								4263.677*** (409.133)
Observations	2780	2780	2780	2780	2780	2780	2780	2780
Adj. R^2	0.552	0.759	0.778	0.805	0.807	0.869	0.869	0.869

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

XRP_{ZAR} stays positive and highly significant in all eight columns (758.1→632.1 as controls accumulate). The Dollar coefficient turns negative once rate blocks are added, consistent with LCU mechanics and South Africa’s rate–FX interactions; VIX is strongly negative without rate controls and weak thereafter. Fit increases from 0.55 to 0.87, while the XRP slope persists—evidence of short-run positive comovement in ZAR terms.

2.6 Dogecoin

Dogecoin Cross-Country synthesis. Across the U.S., UK, Euro Area, China, Brazil, and South Africa, DOGE loadings are uniformly positive and highly significant; Japan and India only exhibit sensitivity when rate spreads are jointly included, with pre-spread specifications still supporting the paper’s central pro-equity comovement claim.

2.6.1 US

Table 41: Dogecoin US Regression

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(\text{DOGE}_{USD})$	16717.286*** (446.882)	14464.520*** (399.297)	15625.078*** (492.253)	15636.854*** (516.144)	14931.635*** (491.326)	11265.437*** (591.995)
$\Delta \text{Commodity}$		-139.632*** (7.022)	-132.599*** (7.195)	-132.777*** (7.567)	-122.208*** (7.206)	-182.402*** (9.080)
ΔDollar			-57.322*** (14.402)	-57.889*** (16.211)	-135.505*** (16.731)	-152.006*** (16.074)
ΔVIX				0.612 (8.033)	25.275*** (7.884)	-25.570*** (9.043)
$\Delta 10\text{Y Yield}$					1268.037*** (110.192)	1843.278*** (119.550)
$\Delta 1\text{M Yield}$						-933.639*** (91.781)
Observations	1091	1091	1091	1091	1091	1091
Adj. R^2	0.562	0.678	0.683	0.682	0.717	0.741

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The DOGE_{USD} coefficient is large, positive, and highly significant in all columns (16.7k in col. 1; 11.3k with the full control set), confirming strong pro-equity comovement in the U.S. Commodities load negatively throughout (cols. 2–6), the Dollar is increasingly negative as macro blocks accumulate (cols. 3–6), and the VIX flips from nil (col. 4) to positive when only levels are in (col. 5) and then negative once short rates are added (col. 6), consistent with discount-rate channels. Long rates are positive, while adding the 1M yield reverses part of that effect (negative), pushing $\text{Adj. } R^2$ up to 0.74 without eroding the core DOGE slope.

2.6.2 UK

Table 42: Dogecoin UK Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{GBP})$	4841.004*** (196.711)	4463.929*** (199.310)	6082.320*** (241.481)	6409.802*** (253.069)	6232.482*** (248.764)	4571.933*** (295.103)	3244.039*** (261.287)	3041.394*** (253.619)
$\Delta \text{Commodity}$		-19.634*** (2.731)	-12.501*** (2.679)	-16.012*** (2.798)	-13.349*** (2.762)	-34.328*** (3.439)	-4.442 (3.304)	-13.531*** (3.358)
ΔDollar			-59.869*** (5.538)	-71.724*** (6.230)	-90.531*** (6.645)	-94.792*** (6.398)	-31.634*** (6.330)	-25.798*** (6.155)
ΔVIX				12.171*** (3.007)	18.113*** (3.057)	-0.438 (3.517)	-13.967*** (3.086)	-19.855*** (3.058)
$\Delta 10Y \text{ Yield}$					300.175*** (42.295)	497.381*** (45.536)	-238.906*** (53.740)	-182.176*** (52.346)
$\Delta 1M \text{ Yield}$						-333.511*** (34.780)	-100.480*** (31.993)	152.561*** (42.274)
$\Delta 10Y \text{ Spread}$							622.401*** (31.270)	663.763*** (30.591)
$\Delta 3M \text{ Spread}$								-323.420*** (36.838)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.357	0.385	0.445	0.452	0.476	0.517	0.646	0.669

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

DOGE_{GBP} is positive and highly significant across all eight columns. Coefficients rise with early global controls (cols. 3–4), then compress as rate blocks enter (cols. 6–8) but remain economically large (3.0–3.2k). Commodities are mostly negative; the Dollar is strongly negative in LCU terms; the VIX flips from positive (cols. 4–5) to negative once short rates/spreads absorb discount-rate risk (cols. 7–8). Term-structure effects are substantial (positive 10Y spread, negative 3M spread), and model fit improves from 0.36 to 0.67 while the core DOGE loading persists.

2.6.3 Eurozone

Table 43: Dogecoin Eurozone Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{EUR})$	310.354*** (12.330)	258.503*** (11.406)	337.104*** (14.209)	335.441*** (15.048)	323.043*** (14.558)	257.354*** (17.575)	265.099*** (16.984)	212.141*** (16.490)
$\Delta \text{Commodity}$		-3.180*** (0.189)	-2.755*** (0.189)	-2.735*** (0.198)	-2.492*** (0.193)	-3.496*** (0.245)	-3.676*** (0.237)	-4.430*** (0.231)
ΔDollar			-3.442*** (0.392)	-3.371*** (0.446)	-5.085*** (0.468)	-5.251*** (0.460)	-3.430*** (0.488)	-1.751*** (0.478)
ΔVIX				-0.072 (0.213)	0.469** (0.213)	-0.420* (0.251)	-0.693*** (0.244)	-1.629*** (0.241)
$\Delta 10Y \text{ Yield}$					27.137*** (2.939)	36.369*** (3.224)	8.174* (4.419)	3.209 (4.161)
$\Delta 1M \text{ Yield}$						-15.817*** (2.462)	-14.352*** (2.382)	17.489*** (3.419)
$\Delta 10Y \text{ Spread}$							48.012*** (5.342)	48.867*** (5.007)
$\Delta 3M \text{ Spread}$								-31.924*** (2.597)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.367	0.498	0.531	0.530	0.564	0.580	0.608	0.656

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

DOGE_{EUR} is positive and precisely estimated in every specification (310 in col. 1; 212 with full controls). Commodities and the Dollar are consistently negative; VIX turns increasingly negative as rate blocks enter (cols. 6–8). Long-rate levels are positive until spreads absorb part of the term-structure signal. Adj. R^2 improves from 0.37 to 0.66, while the DOGE slope remains robust, supporting the pro-equity view in the Euro Area.

2.6.4 Japan

Table 44: Dogecoin Japan Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{JPY})$	460.343*** (15.815)	364.380*** (13.310)	411.923*** (16.784)	369.629*** (17.422)	331.913*** (15.663)	259.104*** (19.195)	74.897*** (15.307)	4.615 (15.144)
$\Delta \text{Commodity}$		-892.074*** (36.531)	-845.966*** (37.574)	-766.101*** (38.358)	-695.530*** (34.388)	-878.473*** (44.414)	-495.946*** (34.881)	-512.327*** (32.370)
ΔDollar			-349.756*** (76.362)	-55.415 (85.073)	-575.573*** (81.707)	-611.429*** (80.466)	242.141*** (65.265)	376.991*** (61.364)
ΔVIX				-296.439*** (41.124)	-130.315*** (37.889)	-289.324*** (44.872)	-432.891*** (33.228)	-557.968*** (32.213)
$\Delta 10Y \text{ Yield}$					8897.666*** (525.775)	10672.853*** (587.421)	463.876 (544.982)	2116.127*** (520.390)
$\Delta 1M \text{ Yield}$						-2819.397*** (444.355)	164.626 (340.060)	679.681** (317.715)
$\Delta 10Y \text{ Spread}$							14312.968*** (468.298)	4396.460*** (862.001)
$\Delta 3M \text{ Spread}$								19915.036*** (1495.392)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.437	0.636	0.643	0.659	0.730	0.739	0.860	0.879

Note: Dependent variable in caption. SE in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

In Japan, DOGE_{JPY} is strongly positive through col. 6 (460→259) but shrinks sharply with spreads: small positive in col. 7 (75, significant) and statistically indistinguishable from zero in col. 8 (4.6, ns). Very large, precisely estimated term-structure coefficients (10Y and 3M spreads) absorb substantial variation, while the Dollar switches sign once those spreads enter (cols. 7–8). Fit rises to 0.88. As with XRP and LTC, spread-heavy JPY specs show sensitivity; the pre-spread columns align with our main claim.

2.6.5 China

Table 45: Dogecoin China Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{CNY})$	162.052*** (9.210)	192.092*** (9.077)	309.262*** (9.717)	342.741*** (9.651)	338.179*** (9.692)	261.988*** (11.560)	224.059*** (13.141)	223.765*** (13.152)
$\Delta \text{Commodity}$		13.467*** (1.166)	18.678*** (1.030)	15.142*** (1.023)	15.582*** (1.025)	6.488*** (1.283)	10.664*** (1.455)	10.593*** (1.460)
ΔDollar			-41.564*** (2.074)	-53.166*** (2.212)	-56.418*** (2.393)	-58.767*** (2.283)	-55.005*** (2.341)	-55.652*** (2.543)
ΔVIX				12.345*** (1.086)	13.385*** (1.122)	5.705*** (1.278)	4.083*** (1.290)	4.166*** (1.296)
$\Delta 10Y \text{ Yield}$					54.295*** (15.691)	142.336*** (16.952)	84.747*** (19.443)	89.659*** (20.857)
$\Delta 1M \text{ Yield}$						-140.761*** (12.918)	-115.797*** (13.439)	-117.081*** (13.586)
$\Delta 10Y \text{ Spread}$							-109.987*** (19.003)	-95.314*** (29.461)
$\Delta 1Y \text{ Spread}$								-19.574 (30.027)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.221	0.305	0.492	0.546	0.550	0.594	0.606	0.606

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

DOGE_{CNY} is positive and highly significant in all columns, rising from 162 (baseline) to 343 with global controls (col. 4), then moderating with rate blocks (224–224 with spreads). Commodities enter positively; the Dollar is strongly negative (LCU mechanics); VIX is positive; and long/short rates are positive (levels) with negative spreads—consistent with discount-rate channels in China. Fit improves from 0.22 to 0.61 while the DOGE coefficient remains robust.

2.6.6 Brazil

Table 46: Dogecoin Brazil Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{BRL})$	8056.630*** (534.779)	5579.006*** (487.983)	7195.354*** (621.519)	7327.600*** (647.746)	6448.737*** (592.233)	2334.214*** (717.282)	2846.505*** (876.999)	2970.987*** (891.306)
$\Delta \text{Commodity}$		-954.556*** (52.506)	-895.963*** (53.997)	-908.761*** (56.808)	-805.526*** (52.139)	-1216.266*** (66.398)	-1208.931*** (66.789)	-1196.098*** (68.763)
ΔDollar			-462.412*** (111.411)	-502.945*** (124.616)	-1253.976*** (123.861)	-1323.915*** (119.352)	-1349.284*** (121.938)	-1384.550*** (129.934)
ΔVIX				43.415 (59.746)	283.495*** (56.649)	-53.309 (65.129)	-29.281 (69.296)	-23.146 (69.745)
$\Delta 10Y \text{ Yield}$					11926.106*** (791.793)	15644.857*** (857.406)	15764.594*** (865.469)	15768.079*** (865.632)
$\Delta 1M \text{ Yield}$						-6274.734*** (664.887)	-6463.863*** (690.485)	-6619.508*** (718.378)
$\Delta 10Y \text{ Spread}$							-296.817 (292.376)	-84.833 (397.614)
$\Delta 1Y \text{ Spread}$								-265.854 (337.877)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.172	0.364	0.373	0.373	0.481	0.520	0.520	0.520

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

DOGE_{BRL} is positive and significant in all columns, though magnitudes decline once rate levels/short rates are added (col. 6) and then stabilize near 2.8–3.0k with spreads. Commodities and the Dollar load strongly negative; VIX turns positive only when rate levels are included (col. 5). Very large, offsetting effects of 10Y and 1M yields underline the importance of discount-rate channels in Brazil. Fit improves from 0.17 to 0.52 while retaining a sizable DOGE slope—consistent with pro-equity behavior.

2.6.7 India

Table 47: Dogecoin India Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{INR})$	227.620*** (8.695)	178.659*** (7.381)	177.681*** (9.157)	181.798*** (9.576)	168.864*** (9.212)	128.922*** (11.563)	-20.167** (9.078)	-18.991** (8.902)
$\Delta \text{Commodity}$		-264.613*** (11.246)	-265.124*** (11.601)	-270.667*** (12.199)	-255.233*** (11.723)	-311.339*** (15.318)	-125.917*** (11.862)	-130.094*** (11.648)
ΔDollar			4.198 (23.225)	-13.163 (26.073)	-126.996*** (27.124)	-142.255*** (26.893)	249.961*** (21.770)	336.355*** (24.966)
ΔVIX				18.909 (12.931)	55.269*** (12.806)	8.170 (15.188)	-90.227*** (10.867)	-93.135*** (10.664)
$\Delta 10Y \text{ Yield}$					1882.725*** (179.567)	2427.025*** (202.165)	343.816** (152.076)	10.310 (157.266)
$\Delta 1M \text{ Yield}$						-869.144*** (155.631)	416.895*** (113.732)	911.007*** (133.867)
$\Delta 10Y \text{ Spread}$							-5978.407*** (173.236)	-3713.135*** (379.650)
$\Delta 2Y \text{ Spread}$								-1904.313*** (285.430)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.386	0.592	0.592	0.593	0.630	0.640	0.828	0.835

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

DOGE_{INR} is positive and significant through col. 6 (228→129) but turns small and negative once 10Y/2Y spreads are added (cols. 7–8). This mirrors the spread-heavy sensitivity seen in JPY: very large term-structure loadings (negative spreads) and a sign-reversing Dollar coefficient suggest over-control/multicollinearity when both levels and multiple spreads are included. Fit rises sharply to 0.83. Pre-spread specifications support our main claim; we flag cols. 7–8 as a robustness check target.

2.6.8 South Africa

Table 48: Dogecoin South Africa Regression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta \log(\text{DOGE}_{\text{ZAR}})$	3150.119*** (136.026)	2978.120*** (141.338)	4532.640*** (160.201)	5055.918*** (159.943)	4945.471*** (159.823)	3449.870*** (182.670)	1563.879*** (124.588)	1563.879*** (124.588)
$\Delta \text{Commodity}$		-186.842*** (45.329)	-10.158 (42.285)	-145.810*** (42.147)	-120.762*** (42.009)	-576.694*** (50.824)	-125.305*** (34.024)	-125.305*** (34.024)
ΔDollar			-1360.470*** (85.170)	-1813.182*** (91.285)	-2001.045*** (98.010)	-2124.423*** (90.814)	-1482.462*** (59.579)	-1482.462*** (59.579)
ΔVIX				479.673*** (44.595)	540.060*** (45.787)	161.430*** (50.291)	218.300*** (31.828)	218.300*** (31.828)
$\Delta 10\text{Y Yield}$					3164.853*** (641.140)	7551.332*** (670.616)	8417.853*** (424.536)	8417.853*** (424.536)
$\Delta 1\text{M Yield}$						-7005.121*** (505.521)	847.142** (374.174)	847.142** (374.174)
$\Delta 10\text{Y Spread}$							-8695.857*** (215.465)	-8695.857*** (215.465)
$\Delta 1\text{Y Spread}$								41488.219*** (986.924)
Observations	1091	1091	1091	1091	1091	1091	1091	1091
Adj. R^2	0.329	0.339	0.464	0.515	0.526	0.597	0.839	0.839

Note: Dependent variable in caption. SE in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

DOGE_{ZAR} remains positive and highly significant across all columns (3.15k baseline; 1.56k with spreads), with strong improvements in fit (Adj. R^2 to 0.84). Dollar coefficients are large and negative, VIX turns positive once macro/rate controls are included, commodities are negative (especially with rate levels), and term-structure coefficients are economically massive. Despite these controls, the DOGE slope persists, consistent with short-run pro-equity exposure in South Africa.

Overall, across six major crypto assets (BTC, ETH, LTC, XRP, USDT, DOGE) and eight equity markets, fully controlled specifications yield predominantly positive and precisely estimated loadings of local-currency crypto returns on local equities. Magnitudes vary by asset and market—largest in US/BRL/ZAR, smaller yet robust in EUR/JPY/INR/CNY—while Tether exhibits a distinct ‘cash-like’ profile with EM-positive/DM-mixed signs. The pattern is consistent with the FX-converted crypto leg sharing the local pricing kernel post-2020 rather than hedging it.

3 Parallel Analysis for Bitcoin Between Pre and Post 2019: Cross Period Evidence and Partial Correlation Interpretation

Overview and Measurement

We study how a BTC shock priced in local currency (LCU) loads into local equities in last section and provide a robustness evidence at all times. So, let's do a parallel analysis for Bitcoin around 2019. before and after 2019. Following the main paper, recall the crypto leg in LCU is

$$r_{i,t}^k \equiv \Delta \log \text{BTC}_{i,t}^{\text{LCU}} = \Delta \log \text{BTC}_t^{\text{USD}} + \Delta \log S_{i,t},$$

where $S_{i,t}$ is LCU per USD. For each market i , we run daily country regressions of LCU equity returns on the LCU BTC leg with progressively richer controls:

$$r_{i,t}^s = \alpha_i + \beta_i r_{i,t}^k + \Gamma_i' X_t + \varepsilon_{i,t},$$

where X_t includes $\Delta \text{Commodity}$, ΔDollar (DXY), ΔVIX , short/long rate changes, and (where applicable) slope/spread terms. We focus on the *fully controlled* specification in each period as the best analog of a *partial* (ceteris-paribus) equity–BTC link.

Interpreting the BTC coefficient as a partial effect. If y and x are standardized and the other regressors are partialled out, the OLS slope on x equals the *partial correlation* $\rho(y, x | X)$. Our daily regressions are not standardized, but the sign and relative magnitude of the fully controlled $\hat{\beta}_i$ across periods provide the same economic information: how strongly equities co-move with BTC *holding fixed* global macro shocks.

Executive Summary (Pre-2019 → Post-2019)

- **BTC–equity linkage attenuates almost everywhere.** The fully controlled BTC slope $\hat{\beta}_i$ declines in the UK, Euro Area, Japan, China, Brazil, India, and South Africa; the US is essentially flat/slightly higher. Sign stays *positive* throughout.
- **Macro channels sharpen.** The Dollar, rate levels, and curve/slopes absorb more variation post-2019; VIX remains strongly negative; commodity effects become more systematic.
- **Model fit rises post-2019.** Adjusted R^2 increases in most markets (UK is the main exception), consistent with macro controls explaining more of daily equity moves while BTC's incremental role is smaller.

Key Cross-Period Comparisons

Tables 49–50 condense the *final-column* estimates from each period’s regression panels (daily data). “Pre” and “Post” denote the pre-2019 and post-2019 samples in the provided PDF tables. Coefficients are read as basis points of equity return per 1% BTC move (i.e., $\beta \times 100$ bps).

Table 49: Fully Controlled BTC Coefficient by Country: Pre vs. Post 2019

	United States	United Kingdom	Euro Area	Japan	China	Brazil	India	South Africa
$\hat{\beta}$ (Pre-2019)	0.049	0.039	0.003	0.003	0.011	0.206	0.002	0.045
$\hat{\beta}$ (Post-2019)	0.055	0.016	0.001	0.002	0.001	0.064	0.001	0.013
Sign change?	+→+	+→+	+→+	+→+	+→+	+→+	+→+	+→+
Direction of shift	<i>flat/slightly up</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>	<i>down</i>

Notes: Entries are the BTC slope $\hat{\beta}$ from each period’s most saturated model (BTC + Commodity + Dollar + VIX + short/long rates + spreads where applicable). Values are expressed as equity bps per 1% BTC move.

Table 50: Adjusted R^2 of Fully Controlled Model: Pre vs. Post 2019

	United States	United Kingdom	Euro Area	Japan	China	Brazil	India	South Africa
Adj. R^2 (Pre-2019)	0.941	0.906	0.915	0.936	0.534	0.815	0.900	0.872
Adj. R^2 (Post-2019)	0.953	0.881	0.937	0.941	0.623	0.821	0.949	0.927

Notes: Values correspond to the final-column specification in each period’s country table.

Country-by-Country Commentary (Fully Controlled Models)

United States. BTC’s partial effect is **essentially unchanged to slightly higher** (0.049→0.055). The Dollar loads more strongly positive post-2019; VIX stays sharply negative. Long rates turn more negative once the front end enters, while short rates are positive—consistent with duration/risk-premium re-pricing dominating at the long end. Fit rises (0.941→0.953). Takeaway: conditional BTC comovement persists, but macro channels carry more of the action.

United Kingdom. BTC falls from 0.039 to 0.016. Commodity remains supportive; Dollar and rate/curve effects strengthen (notably large negatives on the short end and term structure in saturated columns). Adjusted R^2 is slightly lower at the top spec (0.906→0.881), suggesting more idiosyncratic variance in the post period. **Net: conditional BTC linkage weakens materially.**

Euro Area. BTC declines 0.003→0.001. Dollar turns more negative post-2019; 10Y yields are negative when the short end is included, while the front end tends to be positive. Fit improves (0.915→0.937). Equity sensitivity is increasingly explained by USD and rates; **BTC remains a small but positive partial driver.**

Japan. BTC dips (0.003→0.002). Post-2019 rate/curve terms become very large in absolute value, with long-end negatives and front-end positives once the curve is saturated, plus strongly negative VIX. Fit edges up (0.936→0.941). Interpretation: macro integration tightens; **BTC’s incremental same-day footprint shrinks but stays positive.**

China. BTC drops by an order of magnitude (about 0.011→0.001). Dollar turns more negative; long rates become more negative; slope terms gain importance. Fit increases substantially (0.534→0.623). **The conditional BTC signal is now very small, consistent with stronger macro filtering and local market frictions.**

Brazil. BTC shrinks from a very large effect to a still-meaningful one (0.206→0.064). Dollar becomes more negative; front-end rates turn strongly positive while long-end terms go negative in saturated models. Fit improves (0.815→0.821). **BTC remains a positive factor but no longer dominates once macro channels are controlled.**

India. BTC roughly halves (0.002→0.001). Dollar weakens in the richest models; VIX becomes less negative; 10Y tends to flip negative with the curve in, while 1M stays strongly positive; spreads carry significant explanatory power. Fit rises sharply (0.900→0.949). **Result: conditional BTC elasticity is small but robust.**

South Africa. BTC falls by about two thirds (0.045→0.013). Dollar turns strongly negative post-2019; 10Y is negative and 1Y spread strikingly positive in the saturated model; commodity and front-end rates strengthen. Fit increases (0.872→0.927). **BTC’s positive partial effect persists but is much smaller.**

What the Cross-Period Shift Tells Us

Two robust features emerge. First, the *sign* of BTC’s partial effect on equities remains **positive** across markets *in both periods*, aligning with the model’s parity intuition once BTC is translated into LCU. Second, *magnitudes* generally **decline post-2019**, while the macro block—Dollar, VIX, rates, and term structure—absorbs more variation (higher Adj. R^2). This is consistent with a regime in which global macro forces dominate day-to-day equity moves and the incremental same-day BTC signal is correspondingly smaller, even though it remains directionally aligned with equities.

Partial-Correlation Lens (Formal Note)

Let $\tilde{r}_{i,t}^s$ and $\tilde{r}_{i,t}^k$ denote standardized versions of equity and BTC returns after projecting out X_t . In the fully controlled model,

$$\hat{\beta}_i = \frac{\text{Cov}(r_{i,t}^s, r_{i,t}^k | X_t)}{\text{Var}(r_{i,t}^k | X_t)} \implies \text{pcorr}(r_{i,t}^s, r_{i,t}^k | X_t) = \hat{\beta}_i \cdot \frac{\sigma(r_{i,t}^k | X_t)}{\sigma(r_{i,t}^s | X_t)}.$$

I cannot compute partial correlations **without conditional standard deviations**, but comparing $\hat{\beta}_i$ across periods provides the correct *directional* insight: post-2019 partial comovement is generally **weaker**, not stronger, once macro shocks are held fixed.

Bottom Line

Across eight markets, post-2019 daily equity returns display *less* incremental sensitivity to BTC in LCU, while macro factors explain *more*. The equity–BTC link remains **uniformly positive** (sign prediction of the model), but its economic bite is smaller in the later regime.

Table 51: BTC–Equity Partial Correlations (controlling for macro factors): Pre vs. Post 2019

Country	Pre-2019 r	Post-2019 r	Note
United States	0.404	0.912LB	Post uses SE rounded to 0.000; lower bound shown
United Kingdom	0.228	0.543LB	Post uses SE rounded to 0.000; lower bound shown
Euro Area	0.108LB	0.040LB	Both SEs rounded; lower bounds
Japan	0.108LB	0.081LB	Both SEs rounded; lower bounds
China	0.194	0.040LB	Post SE rounded; lower bound
Brazil	0.256	0.791	Exact (SE available in both periods)
India	0.072LB	0.040LB	Both SEs rounded; lower bounds
South Africa	0.261	0.465LB	Post SE rounded; lower bound

Method. For each country and period, I take the BTC coefficient and standard error from the *final* regression column in the PDF tables and compute the partial correlation $r = \sqrt{t^2/(t^2 + \text{df})}$ with $t = \hat{\beta}/\text{SE}$ and $\text{df} = n - p - 1$ (where p counts BTC + Commodity + Dollar + VIX + yield and spread controls used in that column).

Lower bound: The BTC SE was printed as 0.000 (rounded). I conservatively set $\text{SE}_{\max} = 0.0005$, so $t > \hat{\beta}/0.0005$, yielding a minimum (lower-bound) r . True values are *at least* these.

Data. All inputs come directly from the pre- and post-2019 BTC regression PDFs you provided (daily frequency).

Synthesis. Conditioning on macro factors, BTC’s partial link to equity returns is generally *stronger* after 2019, especially in the U.S. and Brazil; the U.K. and South Africa also rise. Continental Europe and Japan remain low, while China and India are small and roughly flat-to-lower. Because several post-2019 BTC SEs are rounded to 0.000, the reported post-period values in those markets are conservative *lower bounds*; the true partial correlations are at least as large.